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MSCI RESEARCH PROJECT

Variational Methods for Option Pricing

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Abstract

Type your abstract here. The abstract is a summary of the contents of the project. It should be brief but informative, and should avoid technicalities as far as possible.

Acknowledgments

Plagiarism statement

The work contained in this thesis is my own work unless otherwise stated.

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Date: April 15, 2026

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Chapter 1

Introduction

1.1 Motivation

Options are derivative contracts that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified strike price K , specified at the time of option purchase. The act of buying or selling the underlying asset is called exercising the option, and is done at a predetermined expiry time T (for European options) or at any time before expiration (for American options). A contract allowing the holder to buy the underlying asset is called a call option, and that allowing the holder to sell the underlying asset is called a put option.

These options are very important financial instruments, and tens of millions of contracts are traded every day just for the S&P 500 index alone, as they are useful for hedging and speculation. For that reason, it is important to be able to price these options accurately. This is a non-trivial task, as the price of an option depends on many factors, such as the underlying asset price which is generally modelled as a stochastic process.

1.2 Literature review

The pioneering work in option pricing was done by Black and Scholes [1], who derived a closed-form solution for the price of the European call option under the assumption that the underlying asset price follows a geometric Brownian motion. This model, known as the Black-Scholes model, is considered the foundational work in option pricing theory. However, it relies on several assumptions such as constant risk-free rate and volatility, which is not always realistic in practice. As a result, Heston [2] derived the European option prices of the proposed stochastic volatility model, allowing the volatility to vary over time to capture more facets of market dynamics.

The American option is much more difficult to price, since it allows the holder to exercise the option at any time before expiration, making it analogous to an optimal stopping problem that does not admit a closed-form solution in general. As a result, various numerical methods have been developed to tackle this problem, such as the binomial tree method [3, 4] for Black-Scholes and Heston respectively, Monte-Carlo simulation methods [5] and more recently variational methods where the optimal stopping problem is formulated as a variational inequality, solved by deep neural networks [6, 7]. This method is adapted to incorporate fractional Sobolev norms to guarantee convergence of the solution to the true option price instead of just pointwise convergence, which has been particularly developed for American option pricing [8].

1.3 Outline of project

Our project focuses on the Black-Scholes and Heston models, for which we will progressively build benchmarks for increasingly complex options. For each model, we begin by deriving the price of a European call option, which serves as a benchmark for the binomial tree method. The binomial tree easily extends to American options, providing a benchmark for the variational neural network method, which we implement with both the standard loss function and the Sobolev loss function. We then extend the analysis to multi-asset options. We first price options on the product of assets, since there exists an equivalent single-asset reduction and can be validated against the benchmarks already established. This foundation allows us to tackle payoffs without such a reduction, such as the maximum or minimum of several assets. Once the numerical results across these methods agree, we will apply our pricing scheme to two further objectives: computing the free boundary of the American option under varying market conditions, and evaluating the Greeks (the sensitivities of the option price to model parameters) to gain insight into exposures and hedging strategies.

Chapter 2

Mathematical Background

2.1 Preliminaries

We will first assemble the necessary mathematical background to understand the option pricing problem and build some foundations into the methods we will be using to solve it. The definitions and results are mostly standard, and can be found in textbooks such as [9, 10] but we will cover them here for self-containment and to set up notation for the rest of the project. As a result, we will only cover the main concepts and not delve too deeply into the details, since our focus is more on the numerical methods and their applications in this project.

2.1.1 Probabilistic Setup

To begin, we require a formalisation of the underlying randomness in our model for the market, which will be done through probability spaces and filtrations. The pricing problem is inherently probabilistic, as we are uncertain about the future evolution of the underlying asset price, thus requiring a rigorous mathematical framework to model this uncertainty.

Definition 2.1.1 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is the sample space, $\mathcal{F}$ is a σ -algebra on Ω , and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure such that $\mathbb{P}(\Omega) = 1$.

The arrival of information through time in the market is modelled by a filtration.

Definition 2.1.2 (Filtration, filtered probability space). A *filtration* on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a non-decreasing family $(\mathcal{F}_t)_{t \geq 0}$ of sub- σ -algebras of $\mathcal{F}$, that is to say, $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for all $s \leq t$. The quadruple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ is called a *filtered probability space*.

We will also define a Martingale, which formalises the idea that future expectations of a process are equal to the current value, which is a key concept in the risk-neutral pricing framework.

Definition 2.1.3 (Martingale). A process $M = (M_t)_{t \geq 0}$ is a *martingale* with respect to the filtration $\{\mathcal{F}_t\}$ and the probability measure $\mathbb{P}$ if $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$.

The Markov assumption is also commonly made in financial modelling and will also be made in our project, which states that the future evolution of the process only depends on the current state and not on the past history.

Definition 2.1.4 (Markov process). A process $X = (X_t)_{t \geq 0}$ is a *Markov process* with respect to the filtration $\{\mathcal{F}_t\}$ if

$$\mathbb{E}[f(X_t) | \mathcal{F}_s] = \mathbb{E}[f(X_t) | X_s] \quad (2.1)$$

for all $0 \leq s \leq t$ and all bounded measurable functions f .

2.1.2 Stochastic processes

The market in practice is not deterministic, so we will mathematically construct our model with stochastic processes.

Definition 2.1.5 (Stochastic process). A *stochastic process* is a family of random variables $\{X_t : t \geq 0\}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

More specifically, we will use Brownian motion to model the underlying asset price dynamics, which is a continuous-time stochastic process with continuous paths and stationary independent increments.

Definition 2.1.6 (Brownian motion). A real-valued process $W = (W_t)_{t \geq 0}$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a *Brownian motion* if it satisfies the following properties:

- $W_0 = 0$ almost surely
- The increment $W_t - W_s$ is normally distributed with mean 0 and variance $t - s$ for all $0 \leq s < t$
- The increments $W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent for all $0 \leq t_0 < t_1 < \dots < t_n$
- The sample paths $t \mapsto W_t(\omega)$ are continuous

The existence of this process is established by Wiener [11] and is very powerful in modelling the randomness in the financial market.

2.1.3 Risk-neutral pricing

We will now formalise the financial market and establish the principle of arbitrage-free valuation that will underpin our option pricing problem.

In our model, the market involves a fixed horizon $T > 0$ (our expiry) and a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$ where $\{\mathcal{F}_t\}$ is the filtration generated by the underlying Brownian motion. The market consists of a risk-free asset B_t that is deterministic and follows the evolution

$$B_t = \exp \left(\int_0^t r(s) ds \right) \quad (2.2)$$

and one or more risky assets S_t that follows an Itô process.

It is important that the market is arbitrage-free, which means that there does not exist any trading strategy that can generate a positive profit with zero initial investment and zero risk. To formalise this, we first define a trading strategy ϕ that is self-financing, which means that that any change in the value of the portfolio is only due to changes in the assets held, without any external cash flow.

Definition 2.1.7 (Self-financing trading strategy). A *self-financing trading strategy* ϕ is a predictable process such that the value of the portfolio $V_t = \phi_t^B B_t + \phi_t^S S_t$ satisfies

$$dV_t = \phi_t^B dB_t + \phi_t^S dS_t \quad (2.3)$$

Definition 2.1.8 (Arbitrage). An *arbitrage* is some trading strategy ϕ such that $V_0 = 0$, $V_T \geq 0$ almost surely and $\mathbb{P}(V_T > 0) > 0$.

By the fundamental theorem of asset pricing, the market is arbitrage-free if and only if there exists a measure $\mathbb{Q}$ such that $\mathbb{Q}$ is equivalent to $\mathbb{P}$ (they have the same null sets) and the discounted price process $\tilde{S}_t = \frac{S_t}{B_t}$ is a martingale under $\mathbb{Q}$.

Given that the market is arbitrage-free, we can price any derivative with payoff $g(\cdot)$ at time t by taking the discounted expectation of the payoff under the risk-neutral measure $\mathbb{Q}$, giving us the risk-neutral pricing formula for European derivative with maturity T :

$$v(t, S) = B_t \mathbb{E}^{\mathbb{Q}} \left[\frac{g(S_T)}{B_T} \mid \mathcal{F}_t \right] \quad (2.4)$$

For this project, we will deal with constant interest rates, so the bond price process reduces to $B_t = e^{rt}$ where r is the constant risk-free rate. Additionally, we will also assume the Markov property (2.1.4). The arbitrage-free price of the European derivative will therefore be given by

$$v(t, S_t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} g(S_T) \mid S_t \right] \quad (2.5)$$

2.1.4 Optimal stopping problem and free boundary

The American option gives the holder the right to exercise the option at any time before maturity, which thus gives us the interpretation of the option price as the solution to an optimal stopping problem.

Definition 2.1.9 (Stopping time). A random variable $\tau : \Omega \rightarrow [0, \infty]$ is a *stopping time* if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$.

The price of the American option can thus be expressed as the $\mathbb{Q}$ -expectation of the discounted payoff at the optimal stopping time τ^* that maximises the expected payoff, giving the optimal stopping problem

$$v(t, S_t) = \sup_{\tau \in [t, T]} \mathbb{E}^{\mathbb{Q}} \left[e^{-r(\tau-t)} g(S_\tau) \mid S_t \right] \quad (2.6)$$

which is simply (2.5) with the additional optimisation over the stopping time τ .

This optimal τ^* is the first time that the option holder chooses to exercise the option, which motivates the notion of a free boundary, which is the boundary separating the exercise region and the continuation region.

Let's work with the case $t = 0$ for simplicity, but the same logic applies for any $t \in [0, T]$. For the American put, the optimal stopping problem (2.6) can be rewritten as

$$v(0, S) = \sup_{\tau \in [0, T]} \mathbb{E} \left[e^{-r\tau} (K - S_\tau)^+ \right] \quad (2.7)$$

which is achieved by an optimal stopping time τ^* taking the form

$$\tau^* = \inf \{ \tau \geq 0 : S_\tau \leq b^*(\tau) \} \quad (2.8)$$

for some optimal exercise boundary $b^*(\tau)$ [5]. This is illustrated in Figure 2.1, where we simulate one realisation of the asset price assuming a geometric Brownian motion and plot it against the exercise boundary, which is found using binary search on the binomial tree solution, where we have

$$b^*(\tau) = \inf\{S : c(\tau, S) > g(S)\} \quad (2.9)$$

where $c(t, S)$ is the continuation value of the option with payoff $g(S)$.

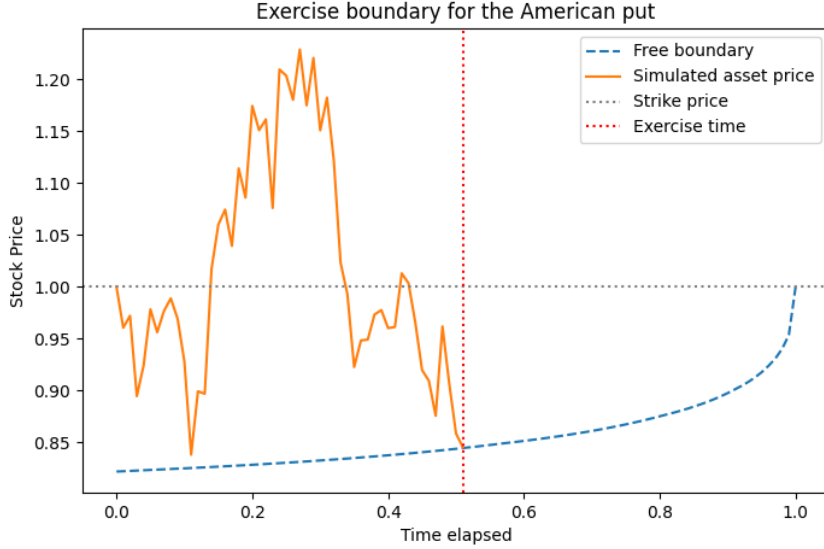


Figure 2.1 Exercise boundary for the American put (adapted from [5])

This formulation describes a hard classification problem, where we classify points in the time-stock price space binarily to be either in the continuation region or the exercise region. However, our numerical solutions to the pricing problem are not exact and there is some level of error in the solution that comes from the Monte Carlo simulations upon the space, the numerical differentiation used in training and the early stopping in the training process. Therefore, it would be more intuitive to perform a soft classification instead, where we assign probabilities to points in the time-stock price space to be in the continuation region or the exercise region, which would give us a more intuitive visualisation of the free boundary and how it evolves over time. We will do this by extending hard classification: that is to say, the continuation region A is

$$A = \{(t, S) : c(t, S) > g(S)\} \cup \{(t, S) : g(S) = 0\} \quad (2.10)$$

where intuitively, the first set is those where the continuation value is greater than the early exercise value, and the second set is those where the payoff is zero, which are at-the-money and out-of-the-money options that are worthless under immediate exercise.

We will consider two methods to extend the definition of the continuation region to a soft classification. The first one is to use a sigmoid function $s : \mathbb{R} \rightarrow (0, 1)$ defined as

$$s(x) = \frac{1}{1 + e^{-x}} \quad (2.11)$$

to smoothly transition the continuation probability from 0 to 1 as the value function transitions

from being below the payoff to being above the payoff. As the sigmoid function is the CDF of the logistic distribution, we can interpret this as modelling the noise in the value function as a logistic distribution. The effect of the noise can be controlled by introducing the softness parameter $\tau > 0$, and our continuation probability function can be defined on the relative difference from the payoff as

$$P_{\text{cont}}(d) = s\left(\frac{d}{\tau}\right), \quad d := \frac{c(t, S) - g(S)}{g(S)} \quad (2.12)$$

To tune the softness parameter, we can define a threshold $I \in (0.5, 1)$ such that we want the continuation probability to be I at some relative difference $\epsilon > 0$, i.e. $P_{\text{cont}}(\epsilon) = I$, which we could solve with some simple algebraic manipulation to get

$$\tau = \frac{\epsilon}{\ln\left(\frac{I}{1-I}\right)} \quad (2.13)$$

The second method is to assume Gaussianity of the noise in our estimate $\hat{c}$ of the continuation value, or in other words

$$\hat{c}(t, S) = c(t, S) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \tilde{\sigma}^2) \quad (2.14)$$

where c is the true (latent) continuation value and $\tilde{\sigma}$ is the standard deviation of the noise, which is unknown. We can then perform n simulations of the solution to get $\hat{c}_1, \dots, \hat{c}_n$ and use the sample mean $\bar{c}$ as our unbiased estimate of the continuation value function, and the sample standard deviation $\hat{\sigma}$ as our estimate. Then $\bar{c}$ is distributed according to a t -distribution with $n - 1$ degrees of freedom, and we can compute the continuation probability as

$$P_{\text{cont}}(t, S) = \mathbb{P}_{t_{n-1}}\left(\frac{\bar{c}(t, S) - g(S)}{\hat{\sigma}(t, S)/\sqrt{n}} > 0\right) \quad (2.15)$$

The caveat with this method is that it requires multiple simulations of the solution, which becomes infeasible for more complex models and higher dimensions, but it gives us a more statistically grounded way to compute the continuation probability.

All that is left is to find a method to obtain the continuation value function, as it is generally latent and not directly observable, given that we predict the American option price that already takes the maximum of the continuation value and the payoff. However, we can use

$$c(t, S) = \mathbb{E}\left[e^{-rh} f(t+h, S_{t+h}) \mid S_t = S\right] \quad (2.16)$$

where $h > 0$ is a small time step and f is the American option price function. We can estimate this expectation by simulating the asset price S_{t+h} at time $t+h$ given $S_t = S$ and evaluating the option price function at the point, then taking the average across all simulations.

2.2 State Space Discretisation

2.2.1 Background

Recall that the option value function $v(t, x)$ is the solution to the optimal stopping problem

$$v(t, S) = \sup_{\tau \in [t, T]} \mathbb{E}\left[e^{-r(\tau-t)} g(S_\tau) \mid S_t = S\right] \quad (2.17)$$

There does not exist a closed-form solution for this problem in general, and thus we have to resort to numerical methods. We first consider methods that discretise the continuous time into a finite grid $\{t_i\}_{i=1}^N$ where we have

$$0 = t_0 < t_1 < \dots < t_N = T \quad (2.18)$$

which we can use to index the underlying state variable S_{t_i} at time t_i . Note that the state variable S_{t_i} could depend on multiple variables, such as in the Heston model, where we have a stochastic variance term encoded in the asset price. The goal is to compute the arbitrage-free price $v(0, S_0)$ at time $t = 0$ given the initial state S_0 . As we know the solution must satisfy the terminal condition $v(T, S) = g(S)$ for all S , we can use backward induction to compute the option value at each time step. This gives rise to the class of numerical methods known as tree methods.

2.2.2 Tree Methods

In a tree model, we discretise the time-state space into a finite grid, where the state variable S_{t_i} can take a finite number of values at each time step given the previous one. We will particularly explore binomial trees in this project, where each state can evolve to one of two possible states at the next time step. Formally, suppose at time t_n we have the state $S_{t_n} \in \{S_n^1, \dots, S_n^{M_n}\}$ where M_n is the number of possible states at time t_n . Then at time t_{n+1} , each state S_n^i can evolve to the state S_{n+1}^j with transition probability

$$p_{ij}^{(n)} = \mathbb{P}(S_{n+1} = S_{n+1}^j \mid S_n = S_n^i) \quad (2.19)$$

This is a discrete-time Markov chain, and to identify the transition probabilities, we can use the risk-neutral condition which states that the discounted price process must be a martingale under the risk-neutral measure. This gives us the condition

$$\sum_{j=1}^{M_{n+1}} p_{ij}^{(n)} S_{n+1}^j = e^{r\Delta t} S_n^i \quad (2.20)$$

where $\Delta t = t_{n+1} - t_n$ and r is the risk-free rate.

With these transition probabilities, we can compute the option value at time t_n given the option value at time t_{n+1} by using backwards induction, which gives us the recursive relation

$$v(t_n, S_n^i) = \max \left(g(S_n^i), e^{-r\Delta t} \sum_{j=1}^{M_{n+1}} p_{ij}^{(n)} v(t_{n+1}, S_{n+1}^j) \right) \quad (2.21)$$

where we take the maximum of the payoff and the discounted expected future price to account for early exercise for the American option. This maximum operator can easily be lifted for all time steps for European options, or only imposed on certain time steps for Bermudan options, leaving the tree method a very flexible method. This backwards induction is applied iteratively starting from the terminal condition at time $t_N = T$ to compute the option value at time $t_0 = 0$.

An issue with tree methods is that the number of states M_n can grow exponentially with the number of time steps N , leading to a very computationally expensive algorithm. To mitigate this, we can use recombining trees where the states at each time step are designed to recombine with each other, so that the number of states M_n grows linearly. Another issue is that, while flexible, the tree method can be difficult to implement for more complex models where the state depends on multiple variables or if we wish to extend it to options operating on multiple assets, but its reliability and interpretability make it a good benchmark for simpler models.

2.3 Physics-Informed Neural Networks

2.3.1 Motivation

Our option pricing problem can alternatively be formulated as a variational inequality which takes the form of a partial differential equation that is independent from the payoff $g(\cdot)$. These PDEs can be of high dimension, and are thus difficult to solve using traditional numerical methods due to the curse of dimensionality. Traditional methods such as finite difference methods require discretisation of the domain and the number of grid points (and hence algorithm complexity) scales exponentially $\mathcal{O}(N^d)$ with the dimension d of the problem.

We can attempt a neural network solution to the PDEs, as neural networks are known to be universal approximators. That is, suppose we have the class of functions $\mathcal{F}$ over a compact set S that can be represented by a neural network with a given architecture, with a continuous target function f^* that we would like to approximate. Then for any target accuracy $\epsilon > 0$, there exists a function $f \in \mathcal{F}$ such that

$$\sup_{x \in S} |f(x) - f^*(x)| \leq \epsilon \quad (2.22)$$

If we have a squashing activation function $\Psi(\cdot)$ that is non-decreasing and satisfies the limits

$$\Psi(-\infty) = 0, \quad \Psi(\infty) = 1 \quad (2.23)$$

then the class of functions $\mathcal{F}$ that can be represented by a neural network with one hidden layer and activation function Ψ is dense in the space of continuous functions on S and thus is a universal approximator [12].

Since the option price function is continuous in the time-asset price domain, we can use a neural network to find an approximation. This is favourable because the time complexity scales with the number of parameters, which is not exponential. To do this, we will use the framework of physics-informed neural networks (PINNs) to solve the PDEs, which have been shown to be effective for solving high-dimensional PDEs [6], by avoiding mesh construction and leveraging the interpolation capability of neural networks [13]. The loss function of the neural network is designed to penalise deviations from the PDE as well as deviations from the boundary conditions. By training the neural network to minimise this loss function, we can obtain an approximation to the solution of the PDE. This method has been applied to option pricing problems in the literature, such as in [14] for the Black-Scholes model, but we will look to apply it to more complex models such as the Heston model as well as to options on multiple assets.

2.3.2 Methodology

We first provide a general setup for PINNs, before modifying it to tailor our specific problem of option pricing. Formally, suppose we have an unknown latent solution $u(t, x)$ which is governed by a PDE of the form

$$\frac{\partial u}{\partial t} + \mathcal{L}[u] = 0, \quad t \in [0, T], x \in \Omega \quad (2.24)$$

where $\mathcal{L}$ is a (not necessarily linear) differential operator. Suppose additionally that this PDE is subject to the boundary conditions

$$u(0, x) = g(x), \quad x \in \Omega \quad (2.25)$$

$$\mathcal{B}[u] = 0, \quad t \in [0, T], x \in \partial\Omega \quad (2.26)$$

where $\mathcal{B}$ is a boundary operator. Then the unknown solution u can be approximated by a deep neural network $f_\theta(t, x)$ where θ are the parameters of the neural network, trained by minimising a loss function composed of deviations from the PDE residual, boundary conditions and initial conditions [15]:

$$L(\theta) = \lambda_r L_r(\theta) + \lambda_{bc} L_{bc}(\theta) + \lambda_{ic} L_{ic}(\theta) \quad (2.27)$$

where the loss components are mathematically defined as

$$L_r(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} |\mathcal{R}_\theta(t_r^i, x_r^i)|^2 \quad (2.28)$$

$$L_{bc}(\theta) = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} |\mathcal{B}[f_\theta](t_{bc}^i, x_{bc}^i)|^2 \quad (2.29)$$

$$L_{ic}(\theta) = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} |f_\theta(0, x_{ic}^i) - g(x_{ic}^i)|^2 \quad (2.30)$$

where the PDE residual $\mathcal{R}_\theta$ is defined as

$$\mathcal{R}_\theta(t, x) = \frac{\partial f_\theta}{\partial t}(t, x) + \mathcal{L}[f_\theta](t, x) \quad (2.31)$$

We additionally use the sampled points $\{x_{ic}^i\}_{i=1}^{N_{ic}}$ from the initial condition domain, $\{(t_{bc}^i, x_{bc}^i)\}_{i=1}^{N_{bc}}$ from the boundary condition domain and $\{(t_r^i, x_r^i)\}_{i=1}^{N_r}$ from the PDE domain, sampled uniformly in their respective domains at each iteration of the gradient descent algorithm.

There are different methods to choose the loss weights $\lambda_r, \lambda_{bc}, \lambda_{ic}$, such as using fixed weights or using adaptive weights that change during training. For simplicity, we will use fixed weights in the project.

For our specific problem of option pricing, we can modify the above setup to fit our variational inequality formulation. The PDE to solve is dependent on the model for the asset dynamics, and the boundary conditions are given by the payoff function of the option. In general, we will have to solve a problem of the form

$$\min \left\{ -\frac{\partial u}{\partial t} + \mathcal{L}[u], u - g \right\} = 0, \quad t \in [0, T], x \in \Omega \quad (2.32)$$

$$u(T, x) = g(x), \quad x \in \Omega \quad (2.33)$$

$$\mathcal{B}_k[u] = 0 \quad t \in [0, T], x \in \partial\Omega, k = 1, \dots, K \quad (2.34)$$

where k indexes the K different boundary conditions. This is very similar to the general setup for PINNs, except that we have a variational inequality instead of a PDE, and we have a terminal condition instead of an initial condition. However, we can still use the framework by noting we can make the substitution $t \mapsto T - t$ to un-negate the negative partial derivative in the PDE term and to convert the terminal condition to an initial condition. The variational inequality can be handled by modifying the PDE residual loss (2.31) to

$$\mathcal{R}_\theta(t, x) = \min \left\{ \frac{\partial f_\theta}{\partial t}(t, x) + \mathcal{L}[f_\theta](t, x), f_\theta(t, x) - g(x) \right\} \quad (2.35)$$

2.4 Sobolev Deep Neural Networks

We can explore a different method to price American puts with neural networks, where this time we work with fractional Sobolev norms to construct our loss function. This is motivated

by the fact that we would be able to guarantee that our neural network converges to the true solution provided that its loss converges instead of just pointwise convergence. Additionally, the loss function components no longer depend on the payoff and is applicable for all situations, a desirable property for a general method.

2.4.1 Preliminaries

The theory of Sobolev spaces are rich and we will not attempt to cover all the details to not divert the attention from the main focus of the project, but we will cover the necessary background to understand the method.

Sobolev Spaces

We first introduce the concept of Sobolev spaces. To begin, we consider those of integer order, defined over an arbitrary domain $\Omega \subset \mathbb{R}^n$, but we will generalise to fractional orders later. These are vector subspaces of various Lebesgue spaces $L^p(\Omega)$, which is the class of all measurable functions u for which

$$\int_{\Omega} |u(x)|^p dx < \infty \quad (2.36)$$

The Sobolev norms are functionals $\|\cdot\|_{m,p}$ where $m \in \mathbb{N}_+$ and $1 \leq p \leq \infty$, defined as follows:

$$\|u\|_{m,p} = \left(\sum_{0 \leq |\alpha| \leq m} \|D^\alpha u\|_p^p \right)^{1/p} \quad 1 \leq p < \infty \quad (2.37)$$

$$\|u\|_{m,\infty} = \max_{0 \leq |\alpha| \leq m} \|D^\alpha u\|_\infty \quad (2.38)$$

where $\|\cdot\|_p$ is the norm in $L^p(\Omega)$. The Sobolev spaces are consequently those on which $\|\cdot\|_{m,p}$ is a norm [16]:

- $H^{m,p}(\Omega) \equiv$ the completion of $\{u \in C^m(\Omega) : \|u\|_{m,p} < \infty\}$ with respect to the norm $\|\cdot\|_{m,p}$
- $W^{m,p}(\Omega) \equiv \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for } 0 \leq |\alpha| \leq m\}$ where $D^\alpha u$ is the weak partial derivative
- $W_0^{m,p}(\Omega) \equiv$ the closure of C_0^∞ in the space $W^{m,p}(\Omega)$

It can be shown by Theorem 3.17 of [16] that for $1 \leq p < \infty$ we have

$$H^{m,p}(\Omega) = W^{m,p}(\Omega) \quad (2.39)$$

so for the rest of this section, we will use the space $H^{m,p}(\Omega)$.

Fractional Sobolev Spaces

Suppose we have a general, possibly nonsmooth, open set $\Omega \subset \mathbb{R}^n$. For any real $s > 0$ and for any $p \in [1, \infty)$, we aim to define the fractional Sobolev spaces $W^{s,p}(\Omega)$.

To construct this, let's first fix the fractional exponent $s \in (0, 1)$. Then for any $p \in [1, +\infty)$ we define $W^{s,p}(\Omega)$ as [17]:

$$W^{s,p}(\Omega) := \left\{ u \in L^p(\Omega) : \frac{|u(x) - u(y)|}{|x - y|^{\frac{n}{p} + s}} \in L^p(\Omega \times \Omega) \right\} \quad (2.40)$$

which can be interpreted as an intermediary Banach space between $L^p(\Omega)$ and $W^{1,p}(\Omega)$ endowed with the norm

$$\|u\|_{W^{s,p}(\Omega)} := \left(\int_{\Omega} |u|^p dx + \int_{\Omega} \int_{\Omega} \frac{|u(x) - u(y)|^p}{|x - y|^{n+sp}} dx dy \right)^{\frac{1}{p}} \quad (2.41)$$

where we have used the *Gagliardo seminorm* of u :

$$[u]_{W^{s,p}(\Omega)} := \left(\int_{\Omega} \int_{\Omega} \frac{|u(x) - u(y)|^p}{|x - y|^{n+sp}} dx dy \right)^{\frac{1}{p}} \quad (2.42)$$

2.4.2 Model

Suppose we would like to price an option on k assets $\mathbf{S} = (S^1, \dots, S^k)$. Similar to before, these assets have theoretically unbounded prices, so we need to truncate it a minimum and maximum price in our model. Thus the domain of the target function can be chosen to be

$$\Omega = (S_{\min}^1, S_{\max}^1) \times \dots \times (S_{\min}^k, S_{\max}^k) \quad (2.43)$$

In our study we will make the simplification that $S_{\min}^i = S_{\min}^j$ and $S_{\max}^i = S_{\max}^j$ for all $i \neq j$, which we can call a and b respectively. Our spaces would consequently have the definitions

$$\Omega = (a, b)^n \quad \Omega_T = (0, T) \times \Omega \quad \Sigma = (0, T) \times \partial\Omega \quad (2.44)$$

where $\partial\Omega =: \Gamma$ is the boundary of Ω , a hyper-rectangle. We will use the representation

$$\Gamma = \bigcup_{i=1}^n (\{x \in [a, b]^n : x_i = a\} \cup \{x \in [a, b]^n : x_i = b\}) \quad (2.45)$$

which is not the simplest, but it is useful as it is decomposed into faces, which is useful when we would like to use the norm derivative later. Note that in the one-dimensional case, this reduces simply to the points $\{a, b\}$.

2.4.3 Loss function

Our loss function would be a weighted sum of four loss components $\sum_{i=1}^4 \lambda_i J_i$, each of which ensures regulation of a specific aspect of our function.

1. The variational loss, which is as before, controlling the accuracy of our function in the interior of our domain:

$$J_1 = \|\min\{\mathcal{G}[f], f - g\}\|_{L^2(\Omega_T), \mu_1}^2 \quad (2.46)$$

2. The energy loss: before we were only controlling the L2 norm of the payoff, but we require the option value to equal the payoff as a function and not just pointwise. Thus we need to incorporate the gradient into the loss so that the spatial slope of the payoff is respected.

$$J_2 = \|f(0, \cdot) - g(\cdot)\|_{H^1(\Omega), \mu_2}^2 \quad (2.47)$$

where we have the norm defined as

$$\|d\|_{H^1(\Omega), \mu_2}^2 = \|d\|_{L^2(\Omega)}^2 + \|\nabla d\|_{L^2(\Omega)}^2 \quad (2.48)$$

and the derivative

$$\nabla d = (\partial_{x_1} d, \dots, \partial_{x_k} d) \quad (2.49)$$

3. J3 (need to explain intuitively)

$$J_3 = \|f(t, x) - g(x)\|_{H^{\frac{3}{4}, \frac{3}{2}}(\Sigma), \mu_3}^2 \quad (2.50)$$

where

$$\|d\|_{H^{\frac{3}{4}, \frac{3}{2}}(\Sigma)}^2 = \|d\|_{H^{0,1}(\Sigma)}^2 + \int_0^T \int_0^T \frac{\|d(t_1, \cdot) - d(t_2, \cdot)\|_{L^2(\Gamma)}^2}{|t_1 - t_2|^{2.5}} dt_1 dt_2 \quad (2.51)$$

$$+ \int_0^T \int_{\Gamma} \int_{\Gamma} \frac{|d(t, x) - d(t, y)|^2}{\|x - y\|^2} d\sigma(x) d\sigma(y) dt \quad (2.52)$$

The time exponent 2.5 is equal to $2p + 1$ where p is the decimal part of the time fractional exponent, and the space exponent 2 is equal to $2\theta + k - 1$ where θ is the decimal part of the space fractional exponent, and k the dimension. Note that here $d\sigma(x)$ and $d\sigma(y)$ indicate the surface measures on Γ , but in practice we can treat these as dx and dy .

4. J4 (need to explain intuitively):

$$J_4 = \|\nabla(f - g) \cdot \nu\|_{H^{\frac{1}{4}, \frac{1}{2}}(\Sigma), \mu_4}^2 \quad (2.53)$$

where the norm derivative is $\nabla(f - g) \cdot \nu = \partial_{x_{i^*}}(f - g)$ given that $x_{i^*} \in \{a, b\}$ and $x_i \in (a, b)$ for $i \neq i^*$. Rigorously, we require the sign but in practice this is not necessary, as we will square in the end. The fractional norm is thus written

$$\|d\|_{H^{\frac{1}{4}, \frac{1}{2}}(\Sigma)}^2 = \|d\|_{L^2(\Sigma)}^2 + \int_0^T \int_0^T \frac{\|d(t_1, \cdot) - d(t_2, \cdot)\|_{L^2(\Gamma)}^2}{|t_1 - t_2|^{1.5}} dt_1 dt_2 \quad (2.54)$$

$$+ \int_0^T \int_{\Gamma} \int_{\Gamma} \frac{|d(t, x) - d(t, y)|^2}{\|x - y\|^2} d\sigma(x) d\sigma(y) dt \quad (2.55)$$

because $H^{0,0} = L^2$.

In practice, we will compute a Monte Carlo estimate of these integrals by sampling pairs (t_1, t_2) from the domain, evaluating the integrand and taking the mean across all pairs. To see this, suppose we can sample iid a random variable X with probability density function $q(x)$. Then for any integrand $\phi(x)$ we have

$$\int \phi(x) dx = \int \frac{\phi(x)}{q(x)} q(x) dx = \mathbb{E} \left[\frac{\phi(X)}{q(X)} \right] \approx \frac{1}{N} \sum_{i=1}^N \frac{\phi(X_i)}{q(X_i)} \quad (2.56)$$

which can be generalised similarly to higher dimensions. By restricting our domain of t_1 and t_2 to be uniform in $[0, 1]$, we have q uniformly equal to 1 and hence we can just evaluate the integrand.

Chapter 3

Black-Scholes Model

The Black-Scholes model [1] is the fundamental model for option pricing, which assumes that the underlying asset price follows a geometric Brownian motion. Here, the market consists of two underlying assets:

- The risk-free asset, described by a deterministic function

$$dB_t = rB_t dt \quad (3.1)$$

where for convenience, we set $B_0 = 1$ and r is the constant risk-free interest rate with continuous compounding. This has the solution $B_t = e^{rt}$ for $t \geq 0$.

- The risky asset S_t , which has the price process of a geometric Brownian motion (GBM), which satisfies the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad 0 \leq t \leq T \quad (3.2)$$

for a given drift μ and volatility σ . We can solve this SDE to yield the unique solution

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right) \quad 0 \leq t \leq T \quad (3.3)$$

Under risk-neutral pricing, we can find an equivalent probability measure $\mathbb{Q}$ under which the discounted stock price is a martingale, and hence all assets earn the risk-free rate in expectation after discounting. Under the risk-neutral measure, the SDE for the risky asset becomes

$$dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}} \quad (3.4)$$

where the drift has been replaced by the risk-free rate r . The derivation can be found in Appendix A.1.

3.1 Closed-form solution for European options

3.2 Binomial Tree Method

[Need to make more cohesive]

To price the put option with the tree method, we will use the binomial discrete-time model. This is an abstraction of the market dynamics, where the stock price can only move to one of

two possible new prices in a time step. This is a simple model which we can make recombining so that the algorithm performs with quadratic time complexity, which will serve as a benchmark for our numerical results for other methods.

To formalise the tree construction, let's fix a time horizon $T > 0$. For some depth $n \geq 1$, divide the interval $[0, T]$ into n even steps each of length $\Delta t = T/n$. Our asset S_t will follow a multiplicative binomial evolution:

$$S_{i+1} = \begin{cases} S_i u & \text{with probability } p \\ S_i d & \text{with probability } 1 - p \end{cases} \quad S_0 > 0 \quad (3.5)$$

where u and d are the model parameters determining the two factors of the stock price change per time step, where we choose $u > 1$ and $d \in (0, 1)$.

As with (3.1), the financial market will also consist of a risk-free asset B_t , discretised so that it satisfies

$$B_{i+1} = B_i e^{r\Delta t} \quad B_0 = 1 \quad (3.6)$$

where r is the continuously-compounded risk-free rate, typically considered interest. For an arbitrage free market, we should additionally ensure that

$$u > e^{r\Delta t} > d \quad (3.7)$$

for sufficiently small Δt , else we can make a riskless profit by shorting the stock and investing in the risk-free asset, or vice versa.

To ensure the absence of arbitrage, we require the discounted asset price to have zero drift (admitting a martingale measure). More precisely, let the discounted price be

$$\tilde{S}_i = e^{-ri\Delta t} S_i \quad (3.8)$$

We require some probability measure $\mathbb{Q}$, equivalent to the physical measure $\mathbb{P}$ (have the same non-zero support) such that

$$\mathbb{E}_{\mathbb{Q}}[\tilde{S}_{i+1} | \mathcal{F}_i] = \tilde{S}_i \quad (3.9)$$

To calculate the risk-neutral probability, we can apply this to our model:

$$e^{-r(i+1)\Delta t} [pS_i u + (1-p)S_i d] = e^{-ri\Delta t} S_i \quad (3.10)$$

Collect the p terms and cancel out the discount factor to get

$$p(S_i u - S_i d) + S_i d = e^{r\Delta t} S_i \quad (3.11)$$

Divide both sides by S_i and rearrange to make p the subject to obtain

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (3.12)$$

which we will use in our binomial model.

We will adopt a backward induction approach to price the American put at time 0, since we already know the price at maturity, which is simply the payoff $g : \mathbb{R}^+ \rightarrow \mathbb{R}$ of the put option:

$$g(S_T) = (K - S_T)^+ \quad (3.13)$$

As the discounted price process is a martingale under the risk-free measure $\mathbb{Q}$, the arbitrage-free value $v(i\Delta t)$ of the derivative at time $i\Delta t$ is necessarily

$$v_i = \underbrace{e^{-r(T-i\Delta t)}}_{\text{discount}} \mathbb{E}_{\mathbb{Q}}[g(S_T) | \mathcal{F}_i] \quad (3.14)$$

We will utilise the Cox-Ross-Rubinstein (CRR) Model [3] for our implementation, which chooses the parameters

$$u = e^{\sigma\sqrt{\Delta t}} \quad d = e^{-\sigma\sqrt{\Delta t}} \quad (3.15)$$

satisfying the arbitrage free condition, and normalising the local variance of returns to the volatility parameter σ^2 . They also conveniently multiply to 1.

To implement this, we need to first compute the price tree. This would be, for the price at step i after j up-moves (so clearly $i \geq j$)

$$S_{i,j} = S_0 u^j d^{i-j} \quad (3.16)$$

As the payoff is known at maturity, we can simply set

$$v_n(j) = g(S_{n,j}) \quad 0 \leq j \leq n \quad (3.17)$$

and recursively set

$$v_i(j) = e^{-r\Delta t} (p^* v_{i+1}(j+1) + (1-p^*) v_{i+1}(j)) \quad (3.18)$$

for $0 \leq j \leq i$. This would be it for European options, but since early exercise is allowed, we will compare this value with the payoff of the price at step i with j up-moves, which is just $f(S_{i,j})$. Thus the value is given by

$$v_i(j) = \max \left(e^{-r\Delta t} (p^* v_{i+1}(j+1) + (1-p^*) v_{i+1}(j)), g(S_{i,j}) \right) \quad (3.19)$$

The root value $v_0(0)$ would be the arbitrage-free option price.

3.3 Physics-Informed Neural Networks

In order to use the framework of PINNs as detailed in section 2.3, we need to first derive the PDE satisfied by the option price, as well as the boundary conditions. We will first work with the one-dimensional case, then discuss the extension to the multi-dimensional case and how we could build a benchmark to against which we evaluate our results.

3.3.1 PINN formulation for one-dimensional case

In order to set up our PINN, we need to first derive the PDE satisfied by the option price. Recall the value function $v : [0, T] \times \mathbb{R}^+ \rightarrow \mathbb{R}$ of an American put with payoff $g : \mathbb{R}^+ \rightarrow \mathbb{R}$ at time t with asset price $S_t = S$ is given by

$$v(t, S) = \sup_{\tau \in [t, T]} \mathbb{E}_t [e^{-r(\tau-t)} g(S_\tau)] \quad (3.20)$$

where τ is any stopping time in $[t, T]$ and $\mathbb{E}_t$ is conditional expectation at time t , given $S_t = S$. For an American option, we always have two options:

- Immediately exercise the option and get payoff $g(S)$. Since this is not necessarily optimal, the payoff must bound the value option from below:

$$v(t, S) \geq g(S) \quad (3.21)$$

- Do nothing and let the option continue to time $t + h$ with $h > 0$ a very small number. By using the supremum definition of $v(t, S)$ (3.20) and swapping the expectation and supremum [18] we arrive at

$$e^{rh}v(t, S) \geq \mathbb{E}_t[v(t + h, S_{t+h})] \quad (3.22)$$

and the inequality turns into an equality at the limit $\delta \rightarrow 0$.

We will additionally assume that the value function $v \in C^{1,2}$ (continuously differentiable with respect to t and twice continuously differentiable with respect to S) so that we can apply Itô's formula. Let's model the asset price S as a geometric Brownian motion satisfying

$$dS_t = rS_t dt + \sigma S_t dW_t \quad (3.23)$$

Then by Ito's lemma and taking expectations of both sides we have

$$\mathbb{E}[v(t + \delta, S_{t+h})] = v(t, S) + \mathbb{E}_t \left[\int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, S_u) + \frac{\partial v}{\partial S}(u, S_u)rS_u + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(u, S_u)\sigma^2 S_u^2 \right) du \right] \quad (3.24)$$

Now let's substitute (3.22) into the LHS, which gives us

$$e^{rh}v(t, S) \geq v(t, S) + \mathbb{E}_t \left[\int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, S_u) + \frac{\partial v}{\partial S}(u, S_u)rS_u + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(u, S_u)\sigma^2 S_u^2 \right) du \right] \quad (3.25)$$

We can use the Taylor approximation $e^{rh} = 1 + rh + o(h)$, which permits us to cancel the $v(t, S)$ on both sides. Dividing both sides by h in the same step, we get

$$rv(t, S) + \frac{o(h)}{h} \geq \mathbb{E}_t \left[\frac{1}{h} \int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, S_u) + \frac{\partial v}{\partial S}(u, S_u)rS_u + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(u, S_u)\sigma^2 S_u^2 \right) du \right] \quad (3.26)$$

Sending $h \rightarrow 0$ makes the higher order term $o(h)/h$ vanish, and the expectation of the integral to collapse to the integrand. Thus we obtain

$$rv(t, S) \geq \frac{\partial v}{\partial t}(t, S) + \frac{\partial v}{\partial S}(t, S)rS + \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(t, S)\sigma^2 S^2 \quad (3.27)$$

or for simplicity, we define the linear parabolic operator $\mathcal{L}$ [8]:

$$\mathcal{L}[v] = rv(t, S) - \frac{\partial v}{\partial t}(t, S)rS - \frac{1}{2} \frac{\partial^2 v}{\partial S^2}(t, S)\sigma^2 S^2 \quad (3.28)$$

Thus we can concisely express this inequality as

$$\min \left(-\frac{\partial v}{\partial t}(t, S) + \mathcal{L}[v], v(t, S) - g(S) \right) \geq 0 \quad (3.29)$$

However, note that at time t , we must either exercise the option or do nothing and continue. One of these two options must be optimal, implying that one of the arguments must be 0. Thus we actually have an equality:

$$\min \left(-\frac{\partial v}{\partial t}(t, S) + \mathcal{L}[v], v(t, S) - g(S) \right) = 0 \quad (3.30)$$

which is the variational equation [19] for the American option. Note this is exactly the formulation of the PINN problem (2.32).

We are now ready to derive the boundary conditions. Our option has a terminal payoff at time T given by $g(S)$, so we have the terminal condition

$$v(T, S) = g(S) \quad (3.31)$$

which satisfies the form of the terminal condition for PINNs (2.33). For the spatial boundary conditions, we have two boundaries to consider: $S \rightarrow 0$ and $S \rightarrow \infty$. In practice, we will have to truncate the domain to $[S_{\min}, S_{\max}]$ for some sufficiently small $S_{\min}$ and sufficiently large $S_{\max}$. At the lower boundary, we have the boundary condition

$$v(t, S_{\min}) = g(S_{\min}) \quad (3.32)$$

because it is always optimal to exercise the option when the asset is at its lowest price. At the upper boundary, we assume the option is too far out of the money to ever finish in the money, and hence the option price is approximately 0. Thus we have the boundary condition

$$v(t, S_{\max}) = 0 \quad (3.33)$$

These three conditions together would form the boundary setup (2.34) for our PINN.

3.3.2 Reduction of multi-asset to one-asset case

Before we apply the PINN framework to the multi-dimensional case, we will first introduce a benchmark for the multi-dimensional case, from which we will build the boundary conditions for the PINN, as they change depending on the payoff function. The benchmark will need to be able to be reduced to a one-dimensional problem, so that we can solve it using the binomial tree method as a reference for our PINN results. This will, in turn, give us confidence that our PINN implementation is correct, and that the results we get for more complex payoffs are reasonable.

In the multi-dimensional case, we have k underlying assets with price processes following a correlated geometric Brownian motion. Mathematically, let each asset S^i satisfy under the risk-free measure $\mathbb{Q}$, for $i = 1, \dots, k$,

$$dS_t^i = rS_t^i dt + \sigma_i S_t^i dW_t^i \quad (3.34)$$

with $\langle W^i, W^j \rangle_t := \mathbb{E}[dW_t^i dW_t^j] = \rho_{ij} dt$ for all i, j pairs. As the geometric Brownian motion is multiplicative, for our benchmark, we will consider the put on the product of the assets. More precisely, we consider the put option with payoff

$$g(S^1, \dots, S^k) = \left(K - \prod_{i=1}^k S^i \right)^+ \quad (3.35)$$

To represent the dynamics of the product of the assets, we can apply Ito's formula to the log of the price process. The log representation is

$$\ln S_t^i = \ln S_0^i + \int_0^t \left(r - \frac{1}{2} \sigma_i^2 \right) ds + \int_0^t \sigma_i dW_s^i \quad (3.36)$$

Define our new asset X to be

$$X_t := \prod_{i=1}^k S_t^i \quad (3.37)$$

and hence we have

$$\ln X_t = \sum_{i=1}^k \ln S_t^i = \ln X_0 + \int_0^t \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) ds + \sum_{i=1}^k \int_0^t \sigma_i dW_s^i \quad (3.38)$$

By taking the differential we have

$$d \ln X_t = \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) dt + \sum_{i=1}^k \sigma_i dW_s^i \quad (3.39)$$

The quadratic variation of the martingale term is

$$\left\langle \sum_{i=1}^k \sigma_i W^i, \sum_{j=1}^k \sigma_j W^j \right\rangle_t = \sum_{i=1}^k \sum_{j=1}^k \sigma_i \sigma_j \rho_{ij} t =: \sigma_X^2 t \quad (3.40)$$

i.e. $\sigma_X^2 = \sum_{i=1}^k \sum_{j=1}^k \sigma_i \sigma_j \rho_{ij}$. So we can express the second term in terms of a new Brownian motion $\widetilde{W}$:

$$\sum_{i=1}^k \sigma_i dW_s^i = d\widetilde{W}_t \quad (3.41)$$

Going back, we have

$$d \ln X_t = \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) dt + \sigma_X d\widetilde{W}_t \quad (3.42)$$

Exponentiating using Ito's formula gets us

$$dX_t = X_t \left[\left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 \right) dt + \sigma_X d\widetilde{W}_t \right] + \frac{1}{2} X_t \sigma_X^2 dt \quad (3.43)$$

which we can rearrange to get

$$dX_t = \left(kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 + \frac{1}{2} \sigma_X^2 \right) X_t dt + \sigma_X X_t d\widetilde{W}_t \quad (3.44)$$

which is a geometric Brownian motion with volatility σ_X and drift r_X where

$$r_X = kr - \frac{1}{2} \sum_{i=1}^k \sigma_i^2 + \frac{1}{2} \sigma_X^2 \quad (3.45)$$

$$= kr + \sum_{i < j} \sigma_i \sigma_j \rho_{ij} \quad (3.46)$$

by matching coefficients. We can verify the equivalence by simulating many price paths for the assets and comparing their product against the simulations of price paths for the equivalent asset. Using the initial values

$$r = 0.05 \quad S_0 = (10, 5, 3)^\top \quad \sigma = (0.2, 0.3, 0.1)^\top \quad R = \begin{pmatrix} 1 & 0.6 & -0.2 \\ 0.6 & 1 & 0.4 \\ -0.2 & 0.4 & 1 \end{pmatrix} \quad (3.47)$$

where R is the correlation matrix, we can plot the histogram of final prices in Figure [3.1].

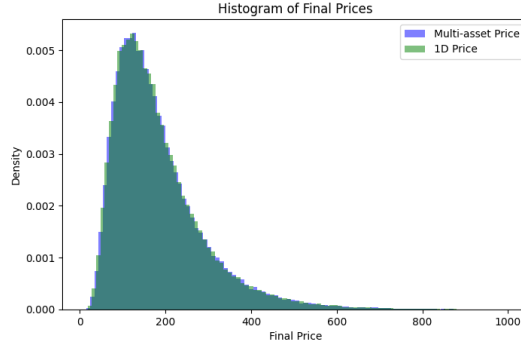


Figure 3.1 Histogram of final prices for product of assets and equivalent asset

They look very similar, and we can further statistically test if they come from the same distribution. Using Kolmogorov-Smirnov's 2-sample test on the null hypothesis that they come from the same distribution, we obtain a p-value of 0.8491, which does not provide sufficient evidence to reject the null hypothesis, which gives us stronger evidence that our derivation is correct.

Therefore, we will be able to price the put option on the product of multiple assets and compare that to the price of the put option on this single constructed asset as a benchmark.

3.3.3 PINN formulation for multi-dimensional case

With the benchmark in place, we now need to derive the variational inequality for the multi-dimensional case where we have k assets. This is a straightforward extension of the one-dimensional case, where the value function $v(t, S^1, \dots, S^k)$ now takes $k+1$ inputs. For notation, we will express the multiple assets as a vector $\mathbf{S} = (S^1, \dots, S^k)^\top$, so the value function is $v(t, \mathbf{S})$. Similarly, the payoff function will be written as $g(S^1, \dots, S^k) = g(\mathbf{S})$.

The value function stays

$$v(t, \mathbf{S}) = \sup_{\tau \in [t, T]} \mathbb{E}_t[e^{-r(\tau-t)} g(\mathbf{S}_\tau)] \quad (3.48)$$

Should we choose not to exercise the option at time t , we can apply the same argument as in the one-dimensional case: considering a small $h > 0$, we can apply Itô's lemma on multiple dimensions to yield

$$\begin{aligned} v(t+h, \mathbf{S}_{t+h}) = v(t, \mathbf{S}) &+ \int_t^{t+h} \left(\frac{\partial v}{\partial t}(u, \mathbf{S}_u) + \sum_{i=1}^k \frac{\partial v}{\partial S^i}(u, \mathbf{S}_u) r S_u^i \right. \\ &\left. + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(u, \mathbf{S}_u) \sigma_i \sigma_j \rho_{ij} S_u^i S_u^j \right) du \end{aligned} \quad (3.49)$$

The steps from here are identical to the one-dimensional case, where we can take expectations of both sides and use the second order Taylor expansion of the exponential to get the variational inequality

$$rv(t, \mathbf{S}) \geq \frac{\partial v}{\partial t}(t, \mathbf{S}) + \sum_{i=1}^k \frac{\partial v}{\partial S^i}(t, \mathbf{S}) r S^i + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(t, \mathbf{S}) \sigma_i \sigma_j \rho_{ij} S^i S^j \quad (3.50)$$

where to simplify into the PINN framework as in (2.32), we can define the operator $\mathcal{L}$ as

$$\mathcal{L}[v] = rv(t, \mathbf{S}) - \sum_{i=1}^k \frac{\partial v}{\partial S^i}(t, \mathbf{S}) r S^i - \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(t, \mathbf{S}) \sigma_i \sigma_j \rho_{ij} S^i S^j \quad (3.51)$$

and the problem reduces to the same variational inequality as in the one-dimensional case (3.30), with a different operator $\mathcal{L}$.

To finish the PINN formulation, we need to derive the boundary conditions. The terminal condition is straightforward, as it is given by the payoff function:

$$v(T, \mathbf{S}) = g(\mathbf{S}) = \left(K - \prod_{i=1}^k S^i \right)^+ \quad (3.52)$$

For the spatial boundary conditions, we will again truncate the domain to

$$\Omega = \left\{ \mathbf{S} = (S^1, \dots, S^k) \in \mathbb{R}^k : S^i \in [S_{\min}^i, S_{\max}^i] \text{ for } i = 1, \dots, k \right\} \quad (3.53)$$

for some sufficiently small and large respective minimum and maximum values for each stock. The lower boundary will be sampled from the set

$$\partial_{\text{lower}} \Omega = \left\{ \mathbf{S} \in \Omega : S^i = S_{\min}^i \text{ for some } i \right\} \quad (3.54)$$

where we have the boundary condition

$$v(t, \mathbf{S}) = g(\mathbf{S}), \quad \mathbf{S} \in \partial_{\text{lower}} \Omega \quad (3.55)$$

and the upper boundary will be sampled from the set

$$\partial_{\text{upper}} \Omega = \left\{ \mathbf{S} \in \Omega : S^i = S_{\max}^i \text{ for some } i \right\} \quad (3.56)$$

with the boundary condition

$$v(t, \mathbf{S}) = 0, \quad \mathbf{S} \in \partial_{\text{upper}} \Omega \quad (3.57)$$

analogous to the one-dimensional case. Of course, this framework could be easily adapted to other payoff functions, where the boundary conditions would change accordingly (but not the variational inequality). But for our benchmark, this is the setup we will use.

3.4 Sobolev Deep Neural Networks

3.5 Numerical Results

3.5.1 Tree benchmark against closed-form solution

First we will evaluate the performance of the benchmark binomial tree method for the one-dimensional case against the closed-form solution, which we would take as the ground truth.

To do this, we will compute the tree option price without taking the maximum of the payoff and continuation value at each step so that we can compare the tree price against the Black-Scholes price.

First, we will evaluate the tree's accuracy on the most sensitive point, which is at time $t = 0$ and when the initial asset price S is equal to the strike price K (at the money). Figure 3.2 shows the convergence of the tree price to the closed-form Black-Scholes price for European put options as we increase the depth of the tree. We can see that around 1000 steps yields a sufficiently close price to the closed-form solution, and further increasing the depth does not yield significant improvements, so this is the depth we will use for our tree benchmark.

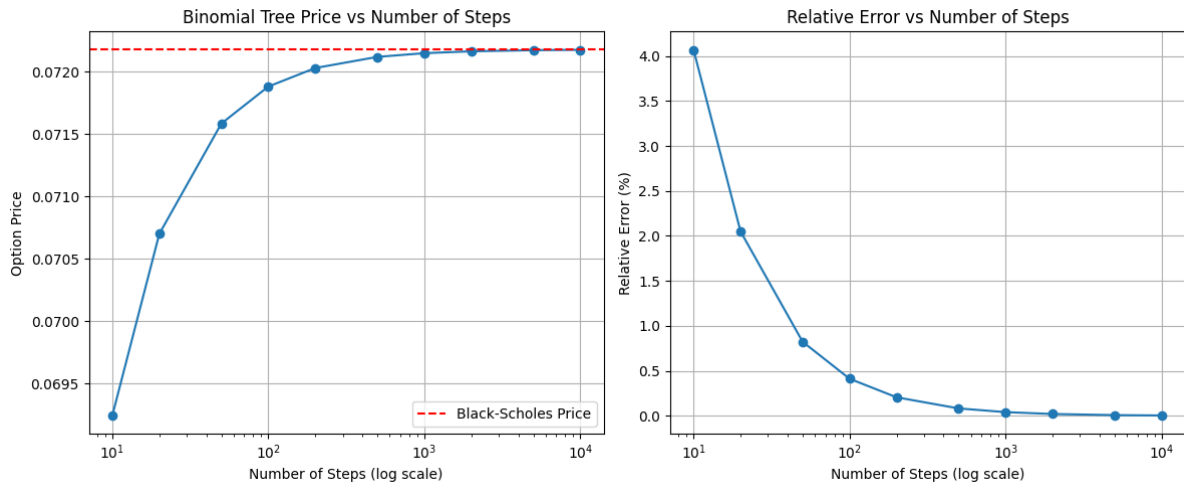


Figure 3.2 Tree convergence at $t = 0, S = K$ over different depths

Evaluating the tree price against the closed-form solution for different initial asset prices S at the initial time $t = 0$ in Figure 3.3, we can see that the tree price is very close to the ground truth, with less than 0.05% of relative error until we are far out of the money where the option price is very close to 0 and unsurprisingly yields greater errors, where machine precision becomes a limiting factor.

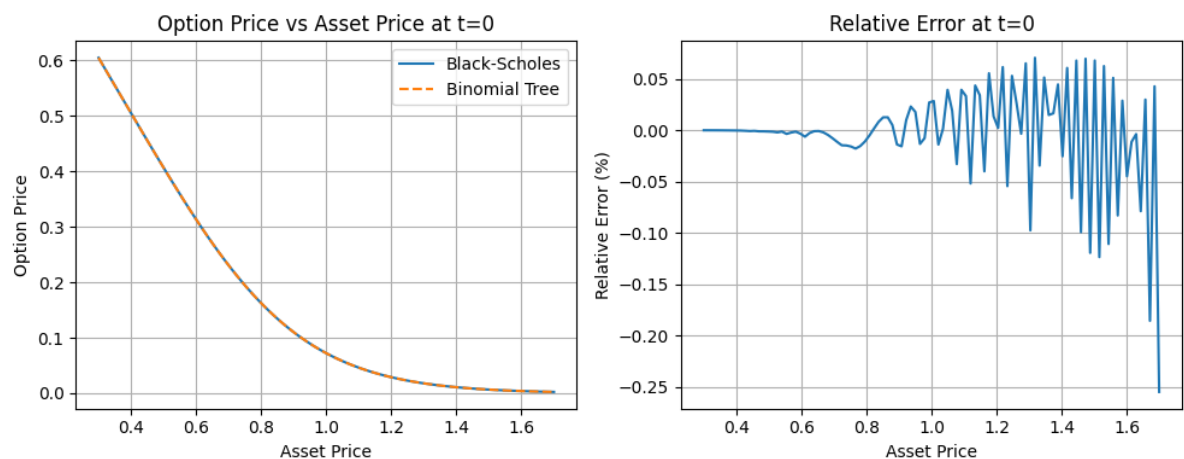


Figure 3.3 Comparison of tree and closed-form prices for European put option with varying initial asset price

We can see that the tree price also closely tracks the term structure of the option price, as shown in Figure 3.4, where the error is again less than 0.05%, decreasing as we get closer to maturity,

which is expected as the tree has not done as many backward-induction steps to get to the initial time, where errors can accumulate.

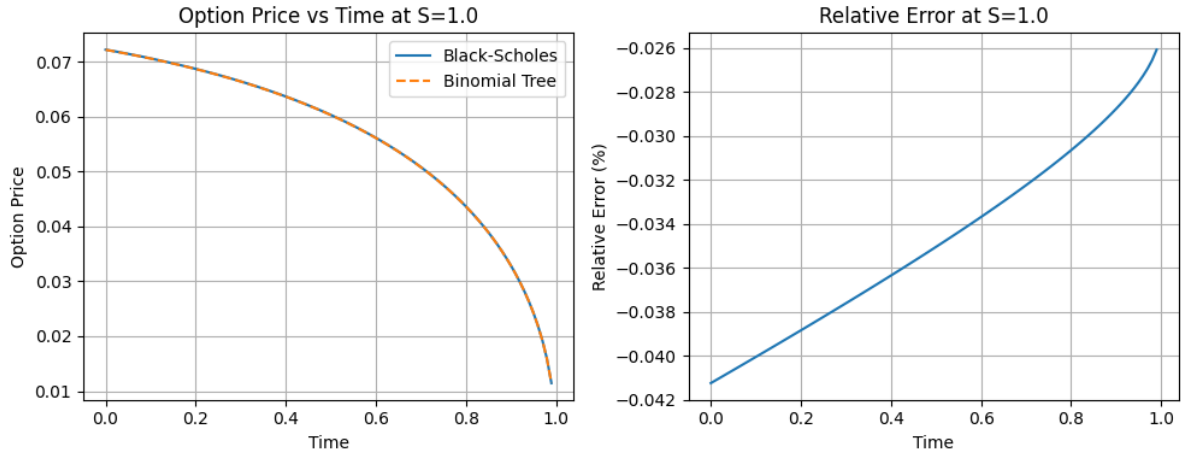


Figure 3.4 Comparison of tree and closed-form prices for European put option with varying time

Overall, we can conclude that the tree benchmark yields very strong results against the closed-form solution, giving us good confidence that it will provide accurate results for American options too, where we do not have a closed-form solution to compare against. This will serve as our benchmark for evaluating the performance of our PINN implementation for American options.

3.5.2 PINN results against tree benchmark

[Do the writeup once all the results are in but put plots in here for now]

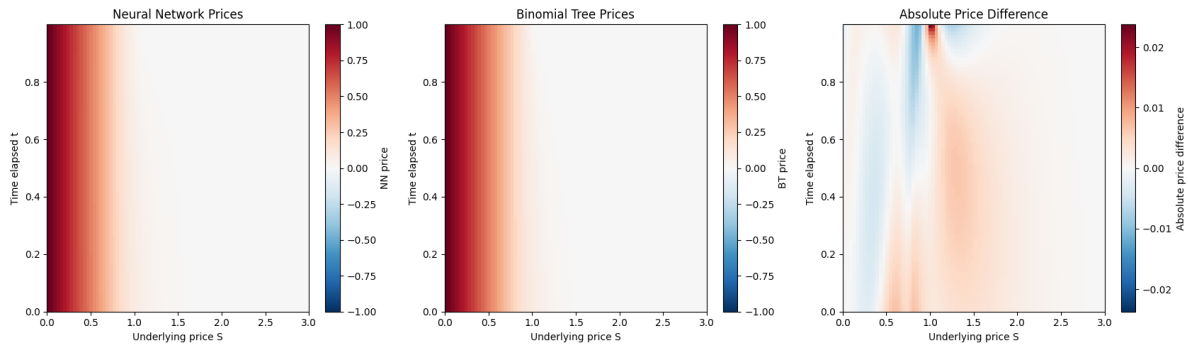


Figure 3.5 PINN vs tree price comparison heatmap across domain

3.5.3 Free boundary

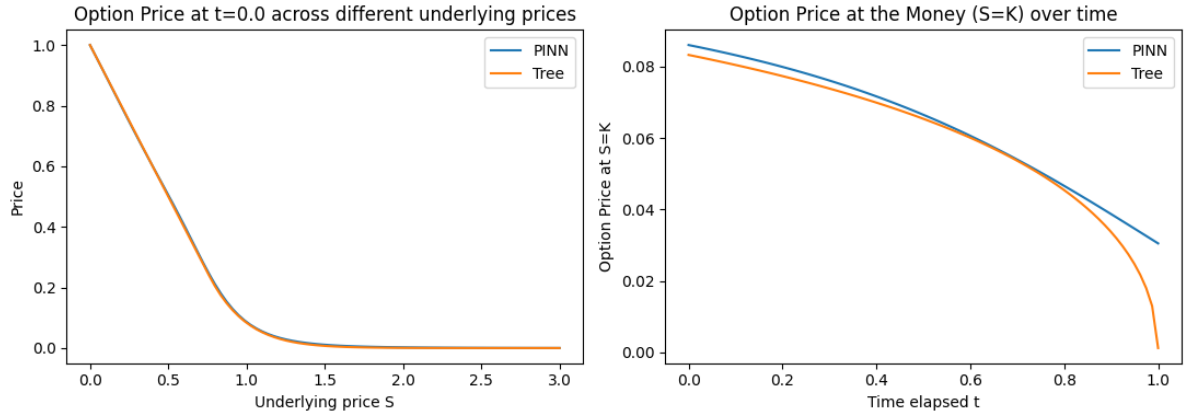


Figure 3.6 Price comparison for varying underlying price, and for term structure at $S = K$

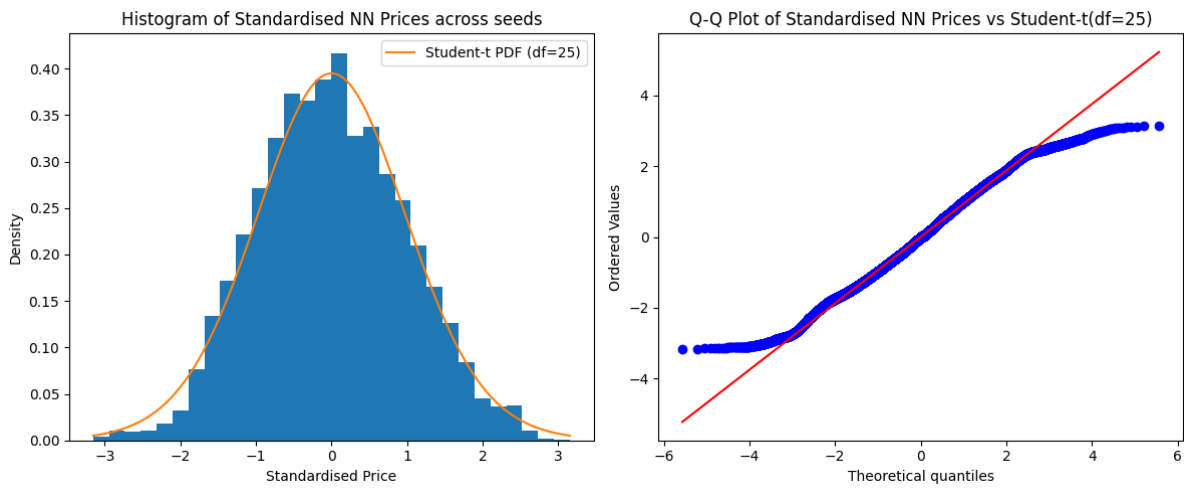


Figure 3.7 Distribution of standardised neural network price estimates across 25 runs

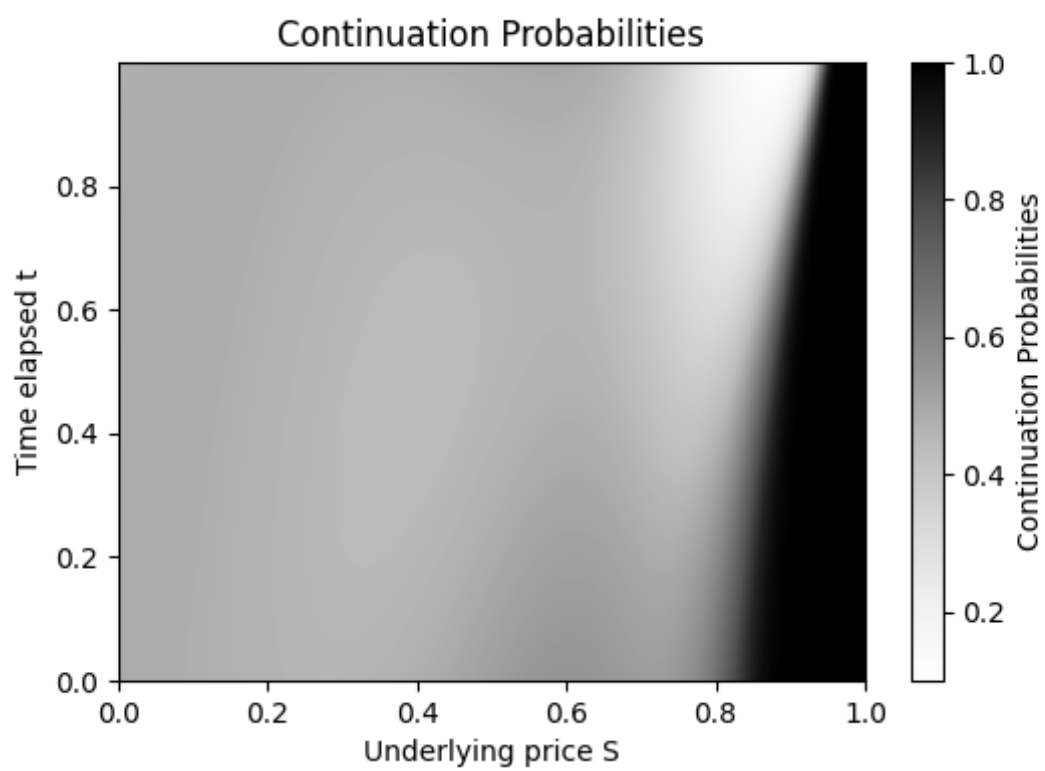


Figure 3.8 Free boundary for PINN

Chapter 4

The Heston Model

The classic Black Scholes model makes the strong assumption that the stock returns are normally distributed with known mean and variance. However, the Black Scholes formula does not depend on the mean, so allowing the mean to vary does not help us to generalise the model. But if we allow volatility to vary according to a stochastic process, we obtain a more generalised model. Heston [2] proposes a model that is not based on the Black Scholes formula that provides a closed-form solution for European call options. In this model, the dynamics under the pricing measure of the asset price S and the variance (squared volatility) process V are governed by the stochastic differential equation system under the risk-neutral measure $\mathbb{Q}$:

$$dS_t = rS_t dt + \sqrt{V_t} dW_t^1 \quad (4.1)$$

$$dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dW_t^2 \quad (4.2)$$

where we have initial conditions $S_0 = s > 0$ and $V_0 = v \geq 0$, and the two Brownian motions are correlated with

$$d\langle W^1, W^2 \rangle_t = \rho dt \quad \rho \in (-1, 1) \quad (4.3)$$

The dynamics of the variance V follow a CIR process with mean reversion rate $\kappa > 0$, long run state $\theta > 0$ and volatility of the volatility $\sigma > 0$. We can enforce the Feller condition $2\kappa\theta \geq \sigma^2$ so that the volatility process stays positive.

4.1 Closed-form solution

There exists a closed-form solution derived by Heston [2] to the price of the European call option. This will serve as a benchmark for our other techniques, which would give us confidence in the accuracy of our other techniques for European options, and by extension, for American options too.

The closed-form solution is derived by using an Ansatz inspired by the Black-Scholes formula, and determining the elements of our guess by substituting it into the PDE we know the solution must satisfy along with the boundary conditions. We will include a sketch of the derivation here which follows those found in [20], but in more detail.

[Derivation]

For expiry $T > 0$, the solution takes the form

$$C(S, V, t) = SP_1 - Ke^{-r(T-t)}P_2 \quad (4.4)$$

where

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left(\frac{e^{-iu \ln K} \phi_j(u)}{iu} \right) du \quad (4.5)$$

for $j = 1, 2$. The characteristic equations $\phi_j(\cdot)$ have the solution

$$\phi_j(u) = \exp(C(T-t; u) + D(T-t; u)V + iu \ln S_0) \quad (4.6)$$

where

$$C(\tau; u) = rui\tau + \frac{a}{\sigma^2} \left\{ (b_j - \rho\sigma ui + d)\tau - 2 \ln \left(\frac{1 - ge^{d\tau}}{1 - g} \right) \right\} \quad (4.7)$$

$$D(\tau; u) = \frac{b_j - \rho\sigma ui + d}{\sigma^2} \left(\frac{1 - e^{d\tau}}{1 - ge^{d\tau}} \right) \quad (4.8)$$

and the quantities

$$g = \frac{b_j - \rho\sigma ui + d}{b_j - \rho\sigma ui - d} \quad (4.9)$$

$$d = \sqrt{(\rho\sigma ui - b_j)^2 - \sigma^2((3-2j)ui - u^2)} \quad (4.10)$$

$$a = \kappa\theta \quad (4.11)$$

$$b_1 = \kappa - \rho\sigma \quad (4.12)$$

$$b_2 = \kappa \quad (4.13)$$

These quantities are derived using Fourier Transforms and complex integration, refer to [20] for more details.

We can plot the call option prices against the stock price S_0 . We expect it to be close to the Black-Scholes formula and for it to collapse to the call option payoff at expiry as the option loses its time value. We see in Figure 4.1 that this behaviour is indeed exhibited.

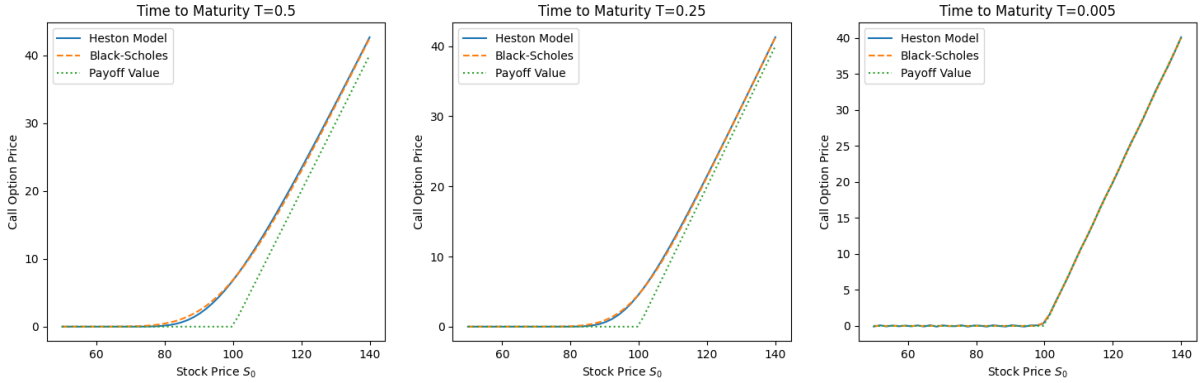


Figure 4.1 Call option values on different maturities

For European options, we can also use put call parity, which is model independent, to obtain the put prices:

$$\text{Call} - \text{Put} = S - Ke^{-rT} \quad (4.14)$$

where we can get the graphs with the same parameters as shown in Figure 4.2. As expected, we also get convergence towards the payoff at smaller time to maturities, and for greater time to maturities, we see that the value of the option is actually lower than the payoff, which is due to the interest rate.

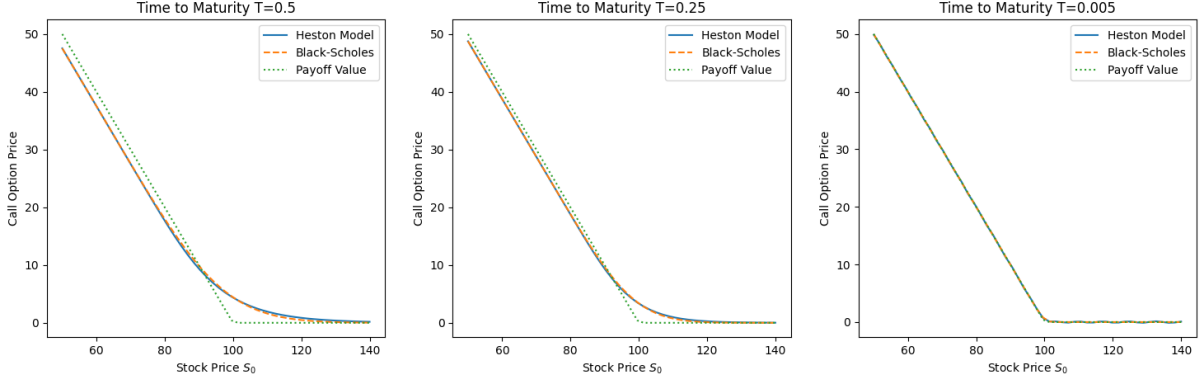


Figure 4.2 Put option values on different maturities

4.2 Tree Solution

Since the closed-form solution is limited to European options, we would like to find ways to approximate solutions to the pricing problem so that we can eventually extend the case to American options, which has no closed-form solution. The first method we will explore is the tree method, where it is easy to modify to the American case by adding an additional comparison to the intrinsic value at each node. We will use the closed-form formula as a benchmark for the tree method for European options, to give us confidence in the solution the tree model provides for American options as it is a simple modification in the algorithm, which we can use as a benchmark for the neural networks we will explore later.

The idea will be similar to the binomial tree we explored earlier in the paper, but we now have two correlated stochastic processes in the Heston model: the asset price process and the variance process. As this tree is not recombining, generating paths for these processes and computing the option prices by backwards induction naively would lead to an exponential time complexity, which is not computationally feasible. Therefore, we will use a grid interpolation approach as in [4] so that the number of nodes grows quadratically in the number of time steps instead.

4.2.1 Algorithm

For the algorithm, we will work under the variance process V and log stock price process Z (which is a simple transformation from S using Itô's formula):

$$dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dW_t^1 \quad (4.15)$$

$$dZ_t = (r - \frac{1}{2}V_t)dt + \sqrt{V_t}dW_t^2 \quad (4.16)$$

where $d\langle W^1, W^2 \rangle_t = \rho$ under the risk-neutral measure $\mathbb{Q}$ and $\kappa, \sigma, \theta, r > 0$ as before. Of course, the value C of the European option with maturity T and payoff function in terms of stock and volatility values $g : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}$ will be

$$C(t, V_t, Z_t) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[g(e^{Z_t}, \sqrt{V_t}) \mid \mathcal{F}_t] \quad (4.17)$$

Euler Scheme

To construct the tree, we need to discretise the process. Suppose we have n steps, then $\Delta t = \frac{T}{n}$. Using the standard Euler scheme, we can have

$$V_{k+1} = V_k + \kappa(\theta - V_k^+) \Delta t + Y_{k+1}^1 \sigma \sqrt{V_k^+ \Delta t} \quad (4.18)$$

$$Z_{k+1} = Z_k + (r - \frac{1}{2} V_k) \Delta t + Y_{k+1}^2 \sqrt{V_k^+ \Delta t} \quad (4.19)$$

where the increment variables (Y_k^1, Y_k^2) are iid in k with probabilities

$$\mathbb{Q}^n(Y_k^1 = i, Y_k^2 = j) = \frac{1}{4}(1 + ij\rho) \quad i, j \in \{-1, 1\} \quad (4.20)$$

We use a different measure $\mathbb{Q}^n$ as it is not the same as the risk-neutral measure in the Heston model. To ensure positivity of the process, we have taken the positive part of the variance process $V_k^+ := \max(0, V_k)$.

Interpolation optimisation

The problem with the naïve Euler scheme is that the tree is not recombining, and hence the number of nodes scales exponentially with increasing time steps, which is not feasible computationally. We will hence, for each time step k , construct a grid of option prices at time k and the backward induction would be a bilinear interpolation of the four nearest points.

To set this up, we need the boundaries of the grid at each time step k , which is given by the minimum and maximum values of V_k and Z_k in the support. In other words, we have

$$z_k^{\max} = \max\{z : \mathbb{Q}^n(Z_k = z) > 0\} \quad (4.21)$$

$$z_k^{\min} = \min\{z : \mathbb{Q}^n(Z_k = z) > 0\} \quad (4.22)$$

$$v_k^{\max} = \max\{v : \mathbb{Q}^n(V_k = v) > 0\} \quad (4.23)$$

$$v_k^{\min} = \min\{v : \mathbb{Q}^n(V_k = v) > 0\} \quad (4.24)$$

$$(4.25)$$

With these boundaries, we have the grid spacing given our mesh sizes $m_z, m_v \in \mathbb{N}^+$:

$$\Delta z_k = (z_k^{\max} - z_k^{\min})/m_z \quad (4.26)$$

$$\Delta v_k = (v_k^{\max} - v_k^{\min})/m_v \quad (4.27)$$

and the gridpoints:

$$\hat{\mathcal{S}}_k = \left\{ \left(v_k^{\min} + i\Delta v_k, z_k^{\min} + j\Delta z_k \right) \mid i \in \{0, \dots, m_v\}, j \in \{0, \dots, m_z\} \right\}$$

We will now define the discretised process $(\tilde{V}, \tilde{Z})$ on the grid space $\hat{\mathcal{S}}_k$. This will involve the Euler scheme functions

$$\nu(y, v, z) = v + \kappa(\theta - v^+) \Delta t + y \sigma \sqrt{v^+ \Delta t} \quad (4.28)$$

$$\zeta(y, v, z) = z + (r - \frac{1}{2} v) \Delta t + y \sqrt{v^+ \Delta t} \quad (4.29)$$

Then using these functions, we can retrieve the next step. We can evolve our process to the next step with

$$(\tilde{V}_{k+1}, \tilde{Z}_{k+1}) = \text{snap}_{Y_{k+1}^3, Y_{k+1}^4}^{\hat{\mathcal{S}}_{k+1}} (\nu(Y_{k+1}^1, \tilde{V}_k, \tilde{Z}_k), \zeta(Y_{k+1}^1, \tilde{V}_k, \tilde{Z}_k)) \quad (4.30)$$

Where $\text{snap}_{i_3, i_4}^{\mathcal{S}} : \mathbb{R}^2 \rightarrow \mathcal{S} \subset \mathbb{R}^2$ is a function mapping $(v, z) \in \mathbb{R}^2$ to one of the grid points in $\mathcal{S} := \mathcal{S}_v \times \mathcal{S}_z$. $i_3, i_4 \in \{0, 1\}$ are additional parameters that determine the corner in which (v, z) is mapped. Visually, this is illustrated in Figure 4.3, where each point in the price-variance space can evolve to four different points in the next time step, each of which we take the interpolation of the four nearest grid points.

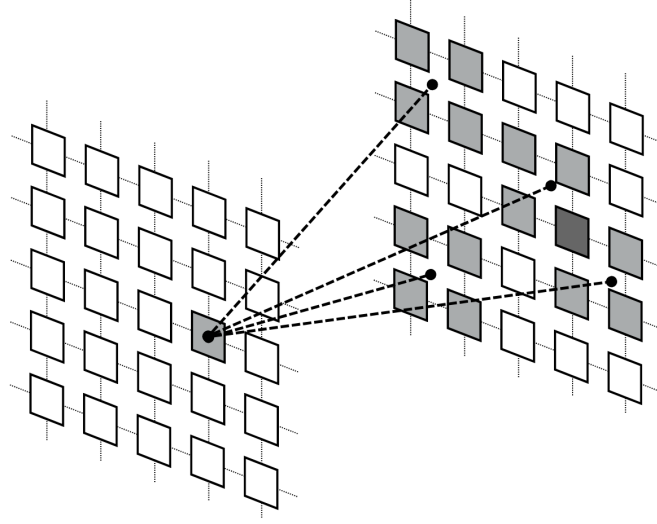


Figure 4.3 Interpolation illustration (adapted from [4])

To do this, we define the 1D snapping functions $\phi_i^{\mathcal{T}} : \mathbb{R} \rightarrow \mathcal{T}$ where we have the one-dimensional grid points $\mathcal{T} \subset \mathbb{R}$ and the snap-right indicator $i \in \{0, 1\}$. The function is defined mathematically as

$$\phi_0^{\mathcal{T}}(u) = \max\{u' \in \mathcal{T} : u' \leq u\} \quad (4.31)$$

$$\phi_1^{\mathcal{T}}(u) = \min\{u' \in \mathcal{T} : u' \geq u\} \quad (4.32)$$

Our 2D snapping function $\text{snap}_{i_3, i_4}^{\mathcal{S}} : \mathbb{R}^2 \rightarrow \mathcal{S}$ where $\mathcal{S} = \mathcal{S}_v \times \mathcal{S}_z \subset \mathbb{R}^2$ is thus

$$(v, z) \mapsto (\phi_{i_3}^{\mathcal{S}_v}(v), \phi_{i_4}^{\mathcal{S}_z}(z)) \quad (4.33)$$

The joint probabilities [4] for Y_{k+1} is given by, for $i_1, i_2 \in \{-1, 1\}$, $i_3, i_4 \in \{0, 1\}$:

$$q_{i_1, i_2, i_3, i_4}(\tilde{V}_k, \tilde{Z}_k) := \tilde{Q}(Y_{k+1}^1 = i_1, Y_{k+1}^2 = i_2, Y_{k+1}^3 = i_3, Y_{k+1}^4 = i_4 \mid (\tilde{V}_k, \tilde{Z}_k)) \quad (4.34)$$

$$= \frac{1}{4} (1 + i_1 i_2 \rho) \times c_{i_3 i_4}^{\hat{\mathcal{S}}_{k+1}} \left(\nu(i_1, \tilde{V}_k, \tilde{Z}_k), \zeta(i_2, \tilde{V}_k, \tilde{Z}_k) \right) \quad (4.35)$$

Here we have used the interpolation quantity [21], where on a general grid set $\mathcal{S} = \mathcal{S}_v \times \mathcal{S}_z \subset \mathbb{R}^2$ we have

$$c_{i_3 i_4}^{\mathcal{S}}(v, z) = \begin{cases} (1 - \tilde{v})(1 - \tilde{z}) & (i_3, i_4) = (0, 0) \\ \tilde{v}(1 - \tilde{z}) & (i_3, i_4) = (1, 0) \\ (1 - \tilde{v})\tilde{z} & (i_3, i_4) = (0, 1) \\ \tilde{v}\tilde{z} & (i_3, i_4) = (1, 1) \end{cases}$$

where we have the difference terms

$$\tilde{v} = \frac{v - \phi_0^{\mathcal{S}_v}(v)}{\phi_1^{\mathcal{S}_v}(v) - \phi_0^{\mathcal{S}_v}(v)} = \frac{v - \phi_0^{\mathcal{S}_v}(v)}{\Delta v_k} \quad (4.36)$$

$$\tilde{z} = \frac{z - \phi_0^{\mathcal{S}_z}(z)}{\phi_1^{\mathcal{S}_z}(z) - \phi_0^{\mathcal{S}_z}(z)} = \frac{z - \phi_0^{\mathcal{S}_z}(z)}{\Delta z_k} \quad (4.37)$$

because our grid is evenly spaced at all time steps k .

Backwards induction

We can now work backwards on the tree, having constructed the grids. Our base case is the final grid, where the price is just the payoff as we are at maturity. For a general time t_k at step k and node $(v, z) \in \hat{\mathcal{S}}_k$, we will generate the four possible steps determined by y_1, y_2 each taking values in $\{-1, 1\}$, and evolve our node with

$$(v, z) \mapsto (\nu(y_1, v, z), \zeta(y_2, v, z)) =: (v', z') \quad (4.38)$$

Which will be in between four points on the grid $\hat{\mathcal{S}}_{k+1}$. These can be found by performing the snap function on our evolved node: for $y_3, y_4 \in \{0, 1\}$ we can encapsulate $y = (y_1, y_2, y_3, y_4)$ and we have

$$(v_y, z_y) = \text{snap}_{y_3, y_4}^{\hat{\mathcal{S}}_{k+1}}(v', z') \quad (4.39)$$

Each of these will have a known option price. Therefore, we can just compute the discounted expectation of these sixteen nodes (two choices for each of $y_1, \dots, y_4$) for the option price at the original node:

$$C(t_k, v, z) = e^{-r\Delta t} \sum_{y \in \mathcal{Y}} q_{y_1, y_2, y_3, y_4}(v, z) \times C(t_{k+1}, v_y, z_y) \quad (4.40)$$

where $\mathcal{Y} = \{-1, 1\}^2 \times \{0, 1\}^2$.

Implementation techniques

If we implemented this method naively, it would be very difficult for us to use the binomial tree as a benchmark because it would need to construct a new tree from scratch for every single value of S_0 , V_0 and T . Thus any heatmaps or plots we create would require hundreds of trees which is not feasible, considering each tree requires many time steps and large mesh sizes to be accurate.

Thus in addition to vectorising our code for each tree, we will instead consider an initial range $[S_0^{\min}, S_0^{\max}]$ and $[V_0^{\min}, V_0^{\max}]$ which we will use to construct our grid bounds. More specifically, $z_k^{\max}$ and $v_k^{\max}$ will be the up-most path starting with $(S_0^{\max}, V_0^{\max})$ instead of simply some individual value (S_0, V_0) , and similar for the down paths. In this way, when we use the binomial tree as a benchmark, we can use any initial value of (S_0, V_0) as long as it's in the original range and compute the price with an interpolation as described above. With this method, we can evaluate across different initial values while constructing just one tree. We sacrifice a little accuracy in doing so because the grids would be more widely spaced, which we can combat by increasing the mesh sizes. Thus, we effectively construct one more complex tree instead of hundreds of simpler ones, which is much more computationally efficient.

4.2.2 Numerical Results

It can be shown that if we have

$$\lim_{n \rightarrow \infty} \Delta t = 0, \quad \lim_{n \rightarrow \infty} \max_{k=1, \dots, n-1} \frac{\Delta v_k^n}{\Delta t_k} = 0, \quad \lim_{n \rightarrow \infty} \max_{k=1, \dots, n-1} \frac{\Delta z_k^n}{\Delta t_k} = 0 \quad (4.41)$$

then the process represented in our Heston tree converges weakly to the original process (V, Z) [4]. Note here we denote Δv_k from before as Δv_k^n to emphasize we have split our maturity T into n steps, and similarly for Δz_k^n .

Taking this into account, we choose our parameters to be $n = 100, m_v = 300, m_z = 600$ which allows our tree to faithfully reproduce the closed form solution while keeping runtime low (after I optimised the algorithm with Python vectorisation). Figure 4.4 are the results after using the parameters $K = 1, T = 1, r = 0.05, \kappa = 2, \theta = 0.04, \sigma = 0.3, \rho = -0.7$. We produce these plots by varying across different values of S_0 (for delta), v_0 (for vega) and time elapsed (for term structure), and all else kept constant.

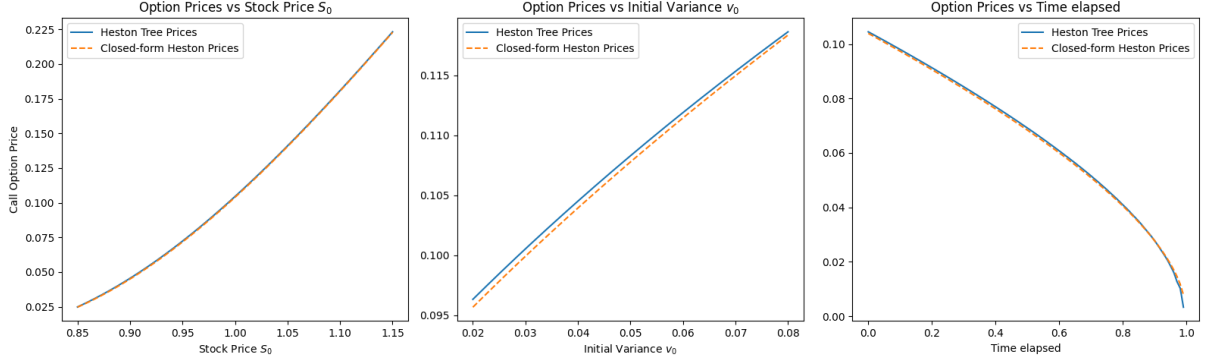


Figure 4.4 Call option prices by the Heston Tree

We can see that the Heston tree does indeed replicate the closed form solution very closely to very small error. This gives us confidence in our implementation, and with an easy modification we can compute the American option prices which we will use as a benchmark.

To compute the American option price with the tree model, all we need to do is to introduce early stopping at every node: we evaluate the intrinsic value the option has at a node and set the option value to be the maximum of the intrinsic value and the discounted continuation value.



Figure 4.5 American vs European option prices for the put

In Figure 4.5 we can see that the European option price indeed bounds the American prices from below, as we would expect. We have confidence that our American prices are accurate since our binomial tree faithfully approximates the closed-form solution for European options. We will now use this as a benchmark for neural network approximations.

4.3 PINN Solution

4.3.1 Variational Equation

We would now like to derive the variational equation for options on assets under the Heston model. Note that for a general process $X_t \in \mathbb{R}^d$ solving the Itô SDE

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t \quad (4.42)$$

where we have the drift $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the volatility $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$ and the m -dimensional Brownian motion W_t . On second-order smooth functions $f \in C^2(\mathbb{R}^d)$, the infinitesimal generator $\mathcal{L}$ is defined as [22]

$$\mathcal{L}f(x) := \lim_{t \rightarrow 0} \frac{\mathbb{E}[f(X_t) | X_0 = x] - f(x)}{t} \quad (4.43)$$

which exists and can be derived with Itô's formula (or Taylor's expansion) to be [23]

$$\mathcal{L}f(x) = \sum_{i=1}^d b_i(x) \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d (\sigma \sigma^\top)_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (4.44)$$

Let's apply this to our Heston model. The drift vector is

$$b(S, V) = \begin{pmatrix} rS \\ \kappa(\theta - V) \end{pmatrix} \quad (4.45)$$

To compute $\sigma \sigma^\top$, we need to represent our correlated Brownian motions in terms of independent ones. Suppose we have independent Brownian motions B^1 and B^2 , we can define

$$W^1 = B^1 \quad (4.46)$$

$$W^2 = \rho B^1 + \sqrt{1 - \rho^2} B^2 \quad (4.47)$$

which has the quadratic variation $d\langle W^1, W^2 \rangle_t = \rho dt$ as required. Using these new Brownian motions, we can rewrite the SDE as

$$d \begin{pmatrix} S_t \\ V_t \end{pmatrix} = \underbrace{\begin{pmatrix} rS_t \\ \kappa(\theta - V_t) \end{pmatrix}}_{b(S_t, V_t)} dt + \underbrace{\begin{pmatrix} S_t \sqrt{V_t} & 0 \\ \rho \sigma \sqrt{V_t} & \sigma \sqrt{V_t} \sqrt{1 - \rho^2} \end{pmatrix}}_{\sigma(S_t, V_t)} \begin{pmatrix} dB_t^1 \\ dB_t^2 \end{pmatrix} \quad (4.48)$$

which means that our covariance matrix is now

$$\sigma(S, V) = \begin{pmatrix} S\sqrt{V} & 0 \\ \rho\sigma\sqrt{V} & \sigma\sqrt{V}\sqrt{1 - \rho^2} \end{pmatrix}. \quad (4.49)$$

which means

$$\sigma \sigma^\top = \begin{pmatrix} S^2 V & \rho \sigma S V \\ \rho \sigma S V & \sigma^2 V \end{pmatrix} \quad (4.50)$$

We now have everything we need for the formula. For any function $f = f(S, V) \in C^2(\mathbb{R}^d)$, we have

$$\mathcal{L}f = rS \frac{\partial f}{\partial S} + \kappa(\theta - V) \frac{\partial f}{\partial V} + \frac{1}{2} \left(S^2 V \frac{\partial^2 f}{\partial S^2} + 2\rho\sigma S V \frac{\partial^2 f}{\partial S \partial V} + \sigma^2 V \frac{\partial^2 f}{\partial V^2} \right) \quad (4.51)$$

This is our infinitesimal generator $\mathcal{L}$ acting on smooth functions f . Suppose we have a European option with payoff $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, then our price function $u(t, S, V)$, the value of the call option with expiry T at time t with $S_t = S$ and $V_t = V$ satisfies

$$\mathcal{G}[u] := ru - \frac{\partial u}{\partial t} + \mathcal{L}u = 0 \quad (4.52)$$

with terminal condition $u(T, S, V) = g(S)$.

As a check, we can set $\kappa = 0$ and $\sigma = 0$ so that $dV = 0$ and volatility becomes a constant, and our model collapses to what we have been working with before. Our generator becomes

$$\mathcal{L}f = rS \frac{\partial f}{\partial S} + \frac{1}{2} S^2 v_0 \frac{\partial^2 f}{\partial S^2} \quad (4.53)$$

which matches our expression from before, since v_0 is the initial variance.

We have derived the variational equation for the European option for simplicity, but it can be easily extended for the American case. With the same argument as ??, the value of the American option must satisfy the PDE

$$\min (\mathcal{G}[u], v(t, S, V) - g(S)) = 0 \quad (4.54)$$

4.3.2 Loss Function

To use a neural network to approximate the function $u(t, S, V)$, we need to establish the relevant boundary conditions. This will help us ensure regularity in all parts of the domain, and we will also incorporate the first derivative to ensure that our neural network approximation also captures the behaviour of the delta (derivative of option price with respect to stock price). The boundary conditions for the call option are provided in [2], but we will adapt them for put options in our study, as they exhibit different behaviour for American options:

1. $t \rightarrow T$: the value at expiry is just the intrinsic value, or the payoff of the put option:

$$u(T, S, V) = \max(0, K - S) \quad (4.55)$$

2. $S \rightarrow 0$: the option value is the strike when the stock is worthless, because we can exercise immediately (for European options we would require a discounting term):

$$u(t, 0, V) = K \quad (4.56)$$

3. $S \rightarrow \infty$: the value of a very far out of the money option is 0, as it is very unlikely for it to expire in of the money:

$$u(t, \infty, V) = 0 \quad (4.57)$$

4. $V \rightarrow 0$: when the volatility is 0, our PDE collapses to

$$rS \frac{\partial u}{\partial S}(t, S, 0) + \kappa \theta \frac{\partial u}{\partial V}(t, S, 0) - ru(t, S, 0) + \frac{\partial u}{\partial t}(t, S, 0) = 0 \quad (4.58)$$

5. $V \rightarrow \infty$: when the volatility is very high, it is almost sure that the product of assets would be zero at some point before expiry, so the value of the option is the strike:

$$u(t, S, \infty) = K \quad (4.59)$$

In practice, as we clearly cannot represent infinity, we will settle for large constants for the asset price $S_{\inf}$ and variance $V_{\inf}$. The minimum for both of these will be taken to be zero. The loss components for our neural network can thus be directly translated. Again, for each epoch, we will take a batch of size B samples for (t, S, V) from our domain

$$\Omega = (0, T) \times (0, S_{\max}) \times (0, V_{\max}) \quad (4.60)$$

and compute the mean squared error like before, where $S_{\max}$ and $V_{\max}$ are chosen constants for our pricing range. Mathematically, let the function represented by the neural network be f . For epoch i , we have the samples $(t_1^{(i)}, \dots, t_B^{(i)})$, $(S_1^{(i)}, \dots, S_B^{(i)})$ and $(V_1^{(i)}, \dots, V_B^{(i)})$, for which we can compute the loss components

$$J_1^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| \min \left\{ \mathcal{G} \left[f \left(t_j^{(i)}, S_j^{(i)}, V_j^{(i)} \right) \right], f \left(t_j^{(i)}, S_j^{(i)}, V_j^{(i)} \right) - \max \left(0, K - S_j^{(i)} \right) \right\} \right\|_2^2 \quad (4.61)$$

$$J_2^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f \left(T, S_j^{(i)}, V_j^{(i)} \right) - \max \left(0, K - S_j^{(i)} \right) \right\|_2^2 \quad (4.62)$$

$$J_3^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f \left(t_j^{(i)}, 0, V_j^{(i)} \right) - K \right\|_2^2 \quad (4.63)$$

$$J_4^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f \left(t_j^{(i)}, S_{\inf}, V_j^{(i)} \right) \right\|_2^2 \quad (4.64)$$

$$J_5^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| r S_j^{(i)} \frac{\partial f}{\partial S}(t_j^{(i)}, S_j^{(i)}, 0) + \kappa \theta \frac{\partial f}{\partial V}(t_j^{(i)}, S_j^{(i)}, 0) - r f(t_j^{(i)}, S_j^{(i)}, 0) + \frac{\partial f}{\partial t}(t_j^{(i)}, S_j^{(i)}, 0) \right\|_2^2 \quad (4.65)$$

$$J_6^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f \left(t_j^{(i)}, S_j^{(i)}, V_{\inf} \right) \right\|_2^2 \quad (4.66)$$

where our final loss will be, again, a convex combination of the six loss components. In other words, for $\lambda \in \Delta_6$, the final loss is

$$J^{(i)} = \lambda^\top J^{(i)} = \sum_{m=1}^6 \lambda_m J_m^{(i)} \quad (4.67)$$

4.3.3 Numerical Results

We are now ready to train the neural network to price our option. We choose to use a learning rate of 1×10^{-3} with the Adam optimiser and a scheduler with step size 2000 and gamma 0.7 and the Sigmoid activation function. By using parameters $K = 1, T = 1, r = 0.1, \kappa = 2, \theta = 0.04, \sigma = 0.3, \rho = -0.7$ obtain the plots as shown in Figure 4.6 for the American put.

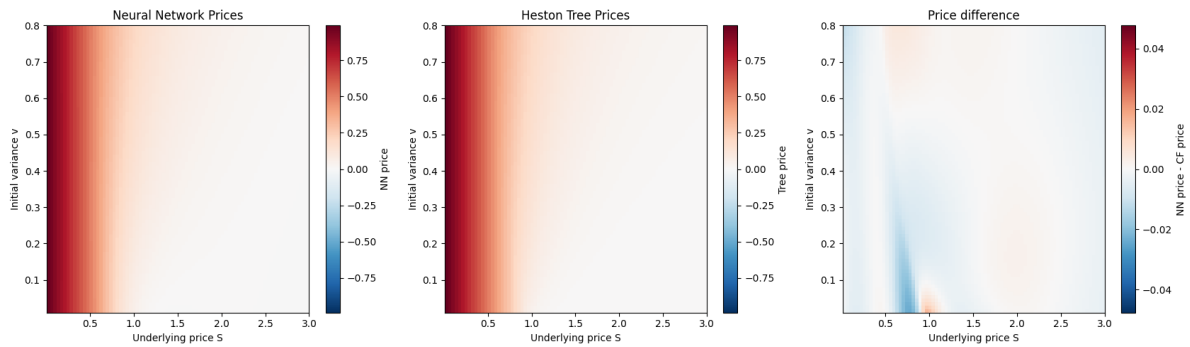


Figure 4.6 American put option prices

We can see that the neural network indeed faithfully reproduces the solution approximated by the binomial tree, even though it tends to perform less well at the money especially at greater

volatilities. This can also be easily shown by fixing a volatility and plotting the approximated option value against the underlying, as shown in in Figure 4.7.

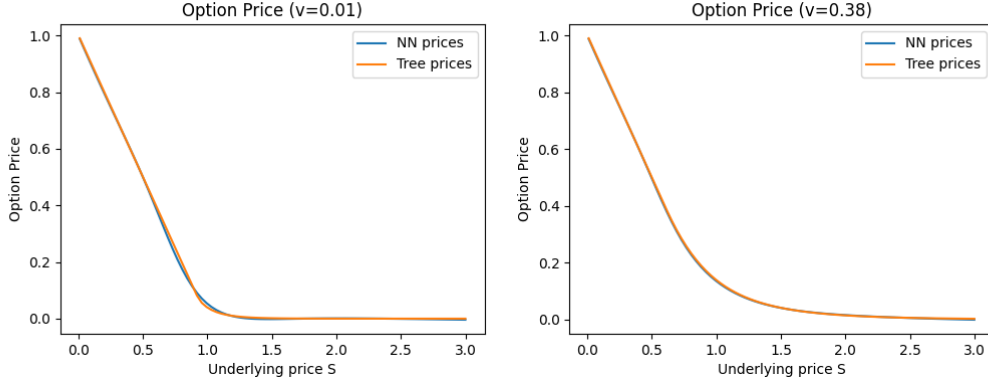


Figure 4.7 American put option prices with volatility fixed

We can see that the tree prices are indeed well approximated by the Neural network, but the at-the-money options are underpriced for higher volatilities. This is due to the lack of a strict high initial variance boundary condition. We can plot the option prices with respect to initial variance in Figure 4.8. Our price does indeed increase with increasing initial variance, but not at the same rate.

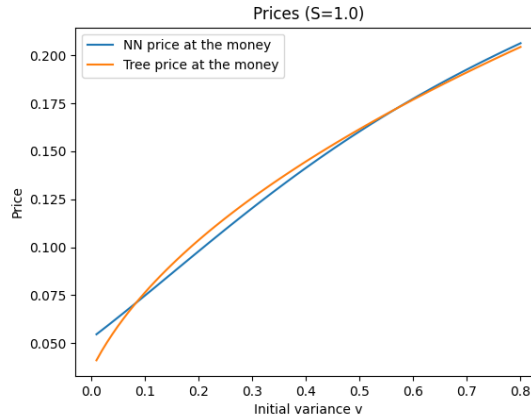


Figure 4.8 At-the-money price with varying initial variance

4.4 Multi-asset PINN

With the neural network established, we are now ready to move on to pricing options on multiple assets. As with the Black-Scholes case, we will price the put on the product of multiple assets because we can find a one-dimensional reduction. Here, we will consider a simplified model. Suppose we have n assets and our assets $S = (S_1, \dots, S_n)^\top$ are governed by the stochastic differential equations:

$$dS = \text{diag}(S)(r dt + \sqrt{V}\sigma dW_1) \quad (4.68)$$

$$dV = \kappa(\theta - V) dt + \bar{\sigma}\sqrt{V} dW_2 \quad (4.69)$$

where we have the vector of iid Brownian motions $W_1 = (W_{1,1}, \dots, W_{1,n})^\top$, variance vector

$\sigma = (\sigma_1, \dots, \sigma_n)^\top$ and the covariance matrix Σ , where for simplicity we will assume all cross-correlations $\Sigma_{ij} = \rho$ for $i \neq j$. Assume additionally that each asset is correlated with the variance process: $dW_{1,j}dW_2 = \rho_j dt$. Note that this means the asset S_i satisfies

$$dS_i = S_i \left(r dt + \sqrt{V} \sigma_i \sum_{j=1}^n \Sigma_{ij} dW_{1,j} \right) \quad (4.70)$$

To guarantee positiveness of the variance process, we will enforce the Feller condition on the parameters

$$2\kappa\theta \geq \bar{\sigma}^2 \quad (4.71)$$

We make these assumptions to allow a one-dimensional reduction for the Heston model, which would provide us a comparable benchmark for our results with the tree method. This would in turn give us confidence for our approximated solutions, should we wish to relax assumptions or to change the payoff.

4.4.1 One-dimensional reduction

Since we would like to find an equivalent formulation of the product of assets $X := \prod_{i=1}^n S_i$, it would be easier to work in the log domain. That is, find the governing SDE for the log process $Y = \log X = \sum_{i=1}^n \log S_i$ and then exponentiate with Itô's lemma to recover the desired result.

Let $Y_i = \log S_i$. Then by Itô's lemma, we have

$$dY_i = \frac{1}{S_i} dS_i - \frac{1}{2S_i} dS_i dS_i \quad (4.72)$$

$$= r dt + \sqrt{V} \sigma_i \sum_{j=1}^n \Sigma_{ij} dW_{1,j} - \frac{1}{2} V \sigma_i^2 dt \quad (4.73)$$

because cross-correlations are zero within W_1 . We can collect terms to get

$$dY = \left(nr - \frac{1}{2} V \sum_{i=1}^n \sigma_i^2 \right) dt + \sqrt{V} \sum_{j=1}^n \sum_{i=1}^n \sigma_i \Sigma_{ij} dW_{1,j} \quad (4.74)$$

For ease of notation, let $A = \sum_{i=1}^n \sigma_i^2$ and $b_j = \sum_{i=1}^n \sigma_i \Sigma_{ij}$. Then we have

$$dY = \left(nr - \frac{1}{2} V A \right) dt + \sqrt{V} \sum_{j=1}^n b_j dW_{1,j} \quad (4.75)$$

Now Y has variance

$$dY dY = V \sum_{j=1}^n b_j^2 dt =: V B dt \quad (4.76)$$

where we let $B := \sum_{j=1}^n b_j^2$ for convenience, we can rewrite dY with a new Brownian motion:

$$dY = \left(nr - \frac{1}{2} V A \right) dt + \sqrt{V B} dW_X \quad (4.77)$$

where we have

$$dW_X = \frac{\sum_{j=1}^n b_j dW_{1,j}}{\sqrt{B}} \quad (4.78)$$

Using our correlation structure, we have

$$b_j = \sigma_j + \rho \sum_{i \neq j} \sigma_i = (1 - \rho) \sigma_j + \rho \underbrace{\sum_{i=1}^n \sigma_i}_{=: S_\sigma} \quad (4.79)$$

So the sum of squares B is just

$$B = \sum_{j=1}^n b_j^2 = \sum_{j=1}^n ((1 - \rho) \sigma_j + \rho S_\sigma)^2 \quad (4.80)$$

$$= (1 - \rho)^2 A + S_\sigma^2 (2\rho(1 - \rho) + n\rho^2) \quad (4.81)$$

We now have all components to exponentiate to recover $X = e^Y$. Using Itô's lemma again, we have

$$dX = X dY + \frac{1}{2} X dY dY \quad (4.82)$$

$$= X \left(nr + \frac{1}{2} V(B - A) \right) dt + X \sqrt{VB} dW_X \quad (4.83)$$

This is close to the Heston model, with the diffusion $\tilde{V} = VB$ which gives us

$$d\tilde{V} = B dV \quad (4.84)$$

$$= \kappa(B\theta - \tilde{V}) dt + \bar{\sigma} \sqrt{B} \sqrt{\tilde{V}} dW_2 \quad (4.85)$$

which is a new CIR process $\tilde{V}$ with parameters $\tilde{\kappa} = \kappa$, $\tilde{\theta} = B\theta$ and $\tilde{\sigma} = \bar{\sigma} \sqrt{B}$. The product of assets nearly gives a direct reduction to a one-dimensional Heston process, but the drift of X contains V which is a stochastic process itself. Thus for simplicity and for our benchmark, we will set $\rho = 0$ which means $B = A$ and the drift term becomes deterministic. As such, we are able to recover a one-dimensional Heston model for the product of assets X with variance $\tilde{V}$ and drift nr .

4.4.2 Variational Equation

Recall for a process $X_t \in \mathbb{R}^d$ solving $dX_t = b(X_t)dt + \sigma(X_t)dW_t$, the infinitesimal generator $\mathcal{L}$ can be derived with Itô's formula [23] on $f(x)$ to be

$$\mathcal{L}f(x) = \sum_{i=1}^d b_i(x) \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d (\sigma \sigma^\top)_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (4.86)$$

In our case, we have a function $f = f(S_1, \dots, S_d, V)$, and we can derive the infinitesimal generator much like the 1D case, but this time we need to sum over our different asset and variance processes. Mathematically, our generator can be expressed to be

$$\mathcal{L}f = \sum_{i=1}^d r S_i \frac{\partial f}{\partial S_i} + \kappa(\theta - V) \frac{\partial f}{\partial V} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \Sigma_{ij} V S_i S_j \frac{\partial^2 f}{\partial S_i \partial S_j} \quad (4.87)$$

$$+ \frac{1}{2} \bar{\sigma}^2 V \frac{\partial^2 f}{\partial V^2} + \sum_{i=1}^d \rho_i \sigma_i \bar{\sigma} V_i S_i \frac{\partial^2 f}{\partial S_i \partial V} \quad (4.88)$$

For any payoff $g(\cdot)$, we have, as before, the variational equation for an European option being

$$\mathcal{G}[f] := rf - \frac{\partial f}{\partial t} + \mathcal{L}u = 0 \quad (4.89)$$

and for an American option, we can introduce the early stopping criterion and the variation equation becomes

$$\min(\mathcal{G}[f], f - g) = 0 \quad (4.90)$$

4.4.3 Loss function

Similar to what we did in the Black-Scholes model, at each epoch we will sample B points to approximate the loss components in each epoch. We would sample for the variance process, asset prices and time elapsed. Introduce our samples, for $\mathbf{S} = (S_1, \dots, S_n)$:

$$t_1, \dots, t_B \in [0, T]^B \quad (4.91)$$

$$V_1, \dots, V_B \in (V_{\min}, V_{\max}) \quad (4.92)$$

$$\mathbf{S}_1, \dots, \mathbf{S}_B \in (S_{\min}^1, S_{\max}^1) \times \dots \times (S_{\min}^n, S_{\max}^n) \quad (4.93)$$

where $S_{\min}^k, S_{\max}^k$ is the modelled minimum and maximum prices asset k could reach, similar for $V_{\min}$ and $V_{\max}$. Of course the variational loss for the American put needs to be included: in epoch i , we have

$$J_1^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| \min \left\{ \mathcal{G} \left[f \left(t_j^{(i)}, \mathbf{S}_j^{(i)}, V_j^{(j)} \right) \right], f \left(t_j^{(i)}, \mathbf{S}_j^{(i)}, V_j^{(j)} \right) - \max \left(0, K - \mathbf{S}_j^{(i)} \right) \right\} \right\|_2^2 \quad (4.94)$$

We will use the same principle ideas for the boundary conditions as the one-dimensional case:

1. $t \rightarrow T$: the value at expiry is just the intrinsic value which is the payoff:

$$u(T, S_1, \dots, S_n, V) = \max \left(0, K - \prod_{k=1}^n S_k \right) \quad (4.95)$$

which means that the loss would be

$$J_2^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f \left(T, \mathbf{S}_j^{(i)}, V_j^{(i)} \right) - \max \left(0, K - \mathbf{S}_j^{(i)} \right) \right\|_2^2 \quad (4.96)$$

2. $S \rightarrow 0$: if one of the assets reaches zero, then the product of assets is consequently zero and we should exercise immediately. Thus the value is the strike. To implement this, let $\tilde{\mathbf{S}}_j^{(i)}$ be $\mathbf{S}_j^{(i)}$ with one of the assets modified manually to 0. Then this loss component is

$$J_3^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f \left(t_j^{(i)}, \tilde{\mathbf{S}}_j^{(i)}, V_j^{(i)} \right) - K \right\|_2^2 \quad (4.97)$$

3. $S \rightarrow \infty$: if one of the assets has a very high price, then the product of assets would also be very high meaning the option would expire worthless out of the money. To implement this, let $\bar{S}_j^{(i)}$ be $S_j^{(i)}$ with one of the assets modified manually to S_{inf} , a large constant. Then this loss component is

$$J_4^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f\left(t_j^{(i)}, \bar{S}_j^{(i)}, V_j^{(i)}\right) \right\|_2^2 \quad (4.98)$$

4. $V \rightarrow 0$: [not done yet]
5. $V \rightarrow \infty$: with infinite volatility, it is almost sure that the intrinsic value of the option would reach the strike at some time before expiry, and thus the value of the option is the strike for the put. To implement this, set V_{inf} a large constant, and we have

$$J_6^{(i)} = \frac{1}{B} \sum_{j=1}^B \left\| f\left(t_j^{(i)}, S_j^{(i)}, V_{\text{inf}}\right) - K \right\|_2^2 \quad (4.99)$$

4.4.4 Numerical results

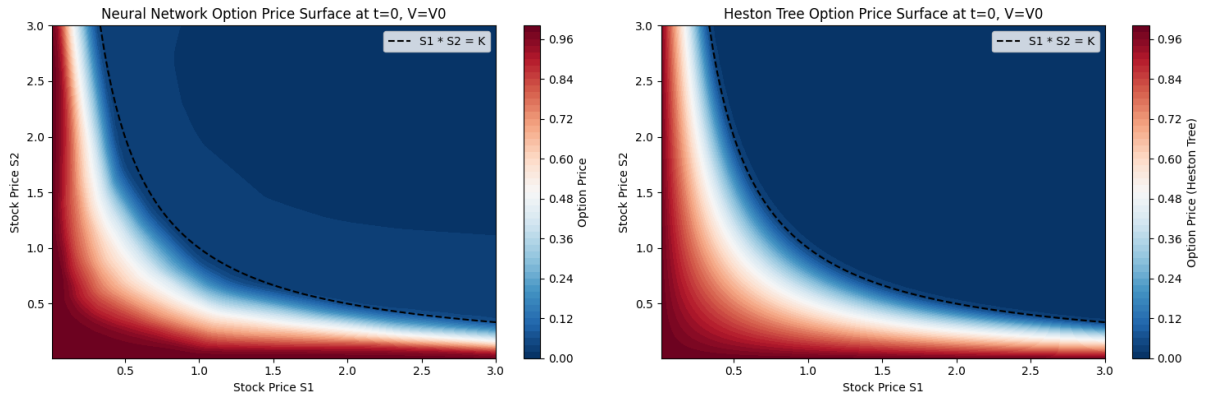


Figure 4.9 Neural network vs 1D reduction tree prices

Appendix A

Appendix 1

A.1 Change of measure for the Black-Scholes model

Consider the discounted stock price

$$\tilde{S}_t := \frac{S_t}{B_t} = e^{-rt} S_t \quad (\text{A.1})$$

and apply Itô's lemma to get a new SDE in terms of the discounted stock price

$$d\tilde{S}_t = \tilde{S}_t[(\mu - r)dt + \sigma dW_t^{\mathbb{P}}] \quad (\text{A.2})$$

which means that under the physical measure $\mathbb{P}$, $\tilde{S}_t$ has nonzero drift unless $\mu = r$, and hence not a martingale. So to enforce no arbitrage in our pricing model, we need a new equivalent probability measure, name it $\mathbb{Q}$, under which $\tilde{S}_t$ is a martingale.

Let's define the quantity

$$c := \frac{\mu - r}{\sigma} \quad (\text{A.3})$$

which measures how much excess drift per unit volatility the stock earns under the physical measure $\mathbb{P}$. Then we can define the Radon-Nikodym derivative

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = Z_t := \exp \left(-cW_t^{\mathbb{P}} - \frac{1}{2}c^2t \right) \quad (\text{A.4})$$

which satisfies Novikov's condition as c is constant so $\mathbb{Q}$ is well defined. The new process

$$W_t^{\mathbb{Q}} := W_t^{\mathbb{P}} + ct \quad (\text{A.5})$$

is a Brownian motion under $\mathbb{Q}$ by Girsanov's theorem. If we substitute this into the original SDE, we obtain

$$\frac{dS_t}{S_t} = \mu dt + \sigma(W_t^{\mathbb{Q}} - cdt) \quad (\text{A.6})$$

$$= (\mu - \sigma c)dt + \sigma dW_t^{\mathbb{Q}} \quad (\text{A.7})$$

$$= rdt + \sigma dW_t^{\mathbb{Q}} \quad (\text{A.8})$$

So now under $\mathbb{Q}$, we have $d\tilde{S}_t = \sigma \tilde{S}_t dW_t^{\mathbb{Q}}$ which has zero drift, and hence $\tilde{S}_t$ is a $\mathbb{Q}$ -martingale. Intuitively, this change of measure reweights possible stock paths so that there is no more risk premium, and all assets earn the risk-free rate in expectation after discounting.

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